

01_Data_EDA

June 23, 2025

1 Data Collection and Exploratory Data Analysis

1.1 Overview

This notebook performs comprehensive data collection from NFL All-22 footage and conducts exploratory data analysis on football formations. We gather real game footage, extract formation features using our fine-tuned YOLOv8 model, and analyze patterns to distinguish between run and pass plays.

1.2 What We're Doing

1.2.1 1. Data Collection

- **YouTube Video Processing:** Download 6 NFL All-22 game videos from YouTube
- **Screenshot Extraction:** Extract specific screenshots at key timestamps using CSV files
- **Automated Processing:** Process multiple videos with timestamp-based frame extraction
- **Storage Management:** Automatically delete large video files after processing to save disk space

1.2.2 2. Feature Extraction

- **Object Detection:** Use our fine-tuned YOLOv8 model to detect player positions in each screenshot
- **Formation Analysis:** Extract comprehensive features from detected formations:
 - Player counts by position (QB, RB, WR, TE, Defense, O-line)
 - Spatial features (formation width, backfield depth, player spacing)
 - Defensive metrics (box defenders, coverage indicators)
- **Multi-Source Processing:** Process images from YouTube screenshots, training data, and test datasets

1.2.3 3. Label Transfer & Enhancement

- **Play Type Classification:** Automatically classify plays as “run” or “pass” based on filename (YouTube Screenshots)
- **Similarity-Based Labeling:** Use formation similarity to predict play types for unlabeled images (training/testing images from Roboflow)
- **Confidence Scoring:** Assign confidence scores to automated predictions

1.2.4 4. Exploratory Data Analysis

- **Statistical Analysis:** Compare feature distributions between run and pass plays
- **Correlation Analysis:** Examine feature relationships within play types
- **Feature Importance:** Identify most discriminative features using Random Forest
- **Visualization:** Create comprehensive plots and heatmaps for pattern discovery

1.3 Key Insights

- **Dataset:** 184 run plays vs 162 pass plays from real NFL footage
- **Predictive Features:** Wide receiver count, formation width, and backfield depth are most important
- **Classification Accuracy:** 66.8% accuracy in predicting run vs pass from formation features
- **Formation Patterns:** Pass formations average 30 pixels wider than run formations

1.4 Output

- **Processed Features:** Comprehensive CSV files with extracted formation features
- **Visual Analysis:** Distribution plots, correlation heatmaps, and feature importance charts
- **Labeled Dataset:** Enhanced dataset with play type labels for machine learning applications

```
[1]: import sys
import os
sys.path.append(os.path.join(os.path.dirname(os.getcwd()))
import logger

from src.data_collection import YouTubeScreenshotCollector
```

```
[2]: # List of dictionaries of video paths and youtube links:
videos = [
    {
        "video_csv_path": "video_csv/Video1.csv",
        "youtube_link": "https://www.youtube.com/watch?v=7ERkAhvmHCo"
    },
    {
        "video_csv_path": "video_csv/Video2.csv",
        "youtube_link": "https://www.youtube.com/watch?v=221V_uAZ1Fg"
    },
    {
        "video_csv_path": "video_csv/Video3.csv",
        "youtube_link": "https://www.youtube.com/watch?v=GDR7QiUjbdw"
    },
    {
        "video_csv_path": "video_csv/Video4.csv",
        "youtube_link": "https://www.youtube.com/watch?v=Pp1jfEqHYxQ"
    },
    {
        "video_csv_path": "video_csv/Video5.csv",
        "youtube_link": "https://www.youtube.com/watch?v=dLbRF7oTl34"
```

```

    },
    {
        "video_csv_path": "video_csv/Video6.csv",
        "youtube_link": "https://www.youtube.com/watch?v=fvaaMcpQBSE"
    },
]

```

```

[3]: # Process all videos in the list
collector = YouTubeScreenshotCollector()

for i, video_info in enumerate(videos):
    print(f"\n--- Processing Video {i+1}/{len(videos)} ---")
    print(f"YouTube Link: {video_info['youtube_link']}")
    print(f"CSV Path: {video_info['video_csv_path']}")

    try:
        # Download video
        video_path = collector.
↳download_youtube_video(video_info['youtube_link'])

        # Load timestamps from CSV
        timestamps, csv_filename = collector.
↳load_timestamps_from_csv(video_info['video_csv_path'])
        print(f"Loaded {len(timestamps)} timestamps from {csv_filename}")

        # Process video and extract screenshots
        num_screenshots = collector.process_video_timestamps(video_path,
↳timestamps, csv_filename)
        print(f"Generated {num_screenshots} screenshots")

        # Delete video to save disk space
        collector.delete_video(video_path)

        print(f" Successfully processed {csv_filename}")

    except Exception as e:
        print(f" Error processing video {i+1}: {e}")
        continue

print(f"\nFinished processing all {len(videos)} videos!")

```

```

--- Processing Video 1/6 ---
YouTube Link: https://www.youtube.com/watch?v=7ERkAhvmHCo
CSV Path: video_csv/Video1.csv
Extracting cookies from chrome
Extracted 3439 cookies from chrome

```

[youtube] Extracting URL: <https://www.youtube.com/watch?v=7ERkAhvmHCo>
 [youtube] 7ERkAhvmHCo: Downloading webpage
 [youtube] 7ERkAhvmHCo: Downloading tv client config
 [youtube] 7ERkAhvmHCo: Downloading player 680f8c75-main
 [youtube] 7ERkAhvmHCo: Downloading tv player API JSON
 Extracting cookies from chrome
 Extracted 3438 cookies from chrome
 [youtube] Extracting URL: <https://www.youtube.com/watch?v=7ERkAhvmHCo>
 [youtube] 7ERkAhvmHCo: Downloading webpage
 [youtube] 7ERkAhvmHCo: Downloading tv client config
 [youtube] 7ERkAhvmHCo: Downloading tv player API JSON
 [info] 7ERkAhvmHCo: Downloading 1 format(s): 18
 [download] Resuming download at byte 185503783
 [download] Destination: /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/data/videos/2024 All 22 Coach film Wk 01 Chiefs vs Ravens.mp4
 [download] 100% of 367.41MiB in 00:00:06 at 55.68MiB/s
 Downloaded video: /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/data/videos/2024 All 22 Coach film Wk 01 Chiefs vs Ravens.mp4
 Loaded 11 timestamps from Video1
 Saved: run_0000_30.jpg
 Saved: pass_0001_88.jpg
 Saved: pass_0002_118.jpg
 Saved: run_0003_146.jpg
 Saved: pass_0004_653.jpg
 Saved: run_0005_683.jpg
 Saved: run_0006_1389.jpg
 Saved: pass_0007_1413.jpg
 Saved: pass_0008_1443.jpg
 Saved: pass_0009_2794.jpg
 Saved: run_0010_3199.jpg
 Generated 11 screenshots
 Deleted video: /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/data/videos/2024 All 22 Coach film Wk 01 Chiefs vs Ravens.mp4
 Successfully processed Video1

--- Processing Video 2/6 ---

YouTube Link: https://www.youtube.com/watch?v=221V_uAZ1Fg
 CSV Path: video_csv/Video2.csv
 Extracting cookies from chrome
 Extracted 3438 cookies from chrome
 [youtube] Extracting URL: https://www.youtube.com/watch?v=221V_uAZ1Fg
 [youtube] 221V_uAZ1Fg: Downloading webpage
 [youtube] 221V_uAZ1Fg: Downloading tv client config
 [youtube] 221V_uAZ1Fg: Downloading tv player API JSON
 Extracting cookies from chrome

Extracted 3438 cookies from chrome
 [youtube] Extracting URL: https://www.youtube.com/watch?v=221V_uAZ1Fg
 [youtube] 221V_uAZ1Fg: Downloading webpage
 [youtube] 221V_uAZ1Fg: Downloading tv client config
 [youtube] 221V_uAZ1Fg: Downloading tv player API JSON
 [info] 221V_uAZ1Fg: Downloading 1 format(s): 18
 [download] Destination: /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/data/videos/2024 All 22 Coaches film Wk 1 Lions Rams.mp4
 [download] 100% of 415.71MiB in 00:00:15 at 26.58MiB/s
 Downloaded video: /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/data/videos/2024 All 22 Coaches film Wk 1 Lions Rams.mp4
 Loaded 11 timestamps from Video2
 Saved: pass_0000_25.jpg
 Saved: pass_0001_55.jpg
 Saved: pass_0002_95.jpg
 Saved: run_0003_695.jpg
 Saved: run_0004_722.jpg
 Saved: run_0005_752.jpg
 Saved: pass_0006_777.jpg
 Saved: run_0007_2632.jpg
 Saved: pass_0008_2672.jpg
 Saved: pass_0009_3295.jpg
 Saved: run_0010_3670.jpg
 Generated 22 screenshots
 Deleted video: /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/data/videos/2024 All 22 Coaches film Wk 1 Lions Rams.mp4
 Successfully processed Video2

--- Processing Video 3/6 ---

YouTube Link: <https://www.youtube.com/watch?v=GDR7QiUjbdw>
 CSV Path: video_csv/Video3.csv
 Extracting cookies from chrome
 Extracted 3438 cookies from chrome
 [youtube] Extracting URL: <https://www.youtube.com/watch?v=GDR7QiUjbdw>
 [youtube] GDR7QiUjbdw: Downloading webpage
 [youtube] GDR7QiUjbdw: Downloading tv client config
 [youtube] GDR7QiUjbdw: Downloading tv player API JSON
 Extracting cookies from chrome
 Extracted 3438 cookies from chrome
 [youtube] Extracting URL: <https://www.youtube.com/watch?v=GDR7QiUjbdw>
 [youtube] GDR7QiUjbdw: Downloading webpage
 [youtube] GDR7QiUjbdw: Downloading tv client config
 [youtube] GDR7QiUjbdw: Downloading tv player API JSON
 [info] GDR7QiUjbdw: Downloading 1 format(s): 18
 [download] Destination: /Users/lmarte/Documents/Projects/CU-

Boulder/Deep_Learning/gridiron-ai/data/videos/2024 All 22 Week 01 Packers vs Eagles.mp4
[download] 100% of 397.26MiB in 00:00:14 at 27.47MiB/s
Downloaded video: /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/data/videos/2024 All 22 Week 01 Packers vs Eagles.mp4
Loaded 11 timestamps from Video3
Saved: run_0000_127.jpg
Saved: run_0001_360.jpg
Saved: pass_0002_540.jpg
Saved: pass_0003_564.jpg
Saved: pass_0004_887.jpg
Saved: run_0005_1281.jpg
Saved: pass_0006_1722.jpg
Saved: pass_0007_2479.jpg
Saved: pass_0008_2518.jpg
Saved: run_0009_2901.jpg
Saved: pass_0010_3996.jpg
Generated 33 screenshots
Deleted video: /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/data/videos/2024 All 22 Week 01 Packers vs Eagles.mp4
Successfully processed Video3

--- Processing Video 4/6 ---

YouTube Link: <https://www.youtube.com/watch?v=Pp1jfEqHYxQ>
CSV Path: video_csv/Video4.csv
Extracting cookies from chrome
Extracted 3438 cookies from chrome
[youtube] Extracting URL: <https://www.youtube.com/watch?v=Pp1jfEqHYxQ>
[youtube] Pp1jfEqHYxQ: Downloading webpage
[youtube] Pp1jfEqHYxQ: Downloading tv client config
[youtube] Pp1jfEqHYxQ: Downloading tv player API JSON
Extracting cookies from chrome
Extracted 3438 cookies from chrome
[youtube] Extracting URL: <https://www.youtube.com/watch?v=Pp1jfEqHYxQ>
[youtube] Pp1jfEqHYxQ: Downloading webpage
[youtube] Pp1jfEqHYxQ: Downloading tv client config
[youtube] Pp1jfEqHYxQ: Downloading tv player API JSON
[info] Pp1jfEqHYxQ: Downloading 1 format(s): 18
[download] Resuming download at byte 83908901
[download] Destination: /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/data/videos/2024 Detroit Lions vs Tampa Bay Buccaneers Week 2.mp4
[download] 100% of 384.79MiB in 00:00:26 at 14.58MiB/s
Downloaded video: /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/data/videos/2024 Detroit Lions vs Tampa Bay Buccaneers Week 2.mp4

Loaded 11 timestamps from Video4
Saved: pass_0000_26.jpg
Saved: pass_0001_51.jpg
Saved: pass_0002_106.jpg
Saved: pass_0003_503.jpg
Saved: pass_0004_1052.jpg
Saved: run_0005_1253.jpg
Saved: run_0006_1282.jpg
Saved: pass_0007_1311.jpg
Saved: pass_0008_1857.jpg
Saved: pass_0009_1884.jpg
Saved: run_0010_2822.jpg
Generated 44 screenshots
Deleted video: /Users/lmarte/Documents/Projects/CU-
Boulder/Deep_Learning/gridiron-ai/data/videos/2024 Detroit Lions vs Tampa Bay
Buccaneers Week 2.mp4
Successfully processed Video4

--- Processing Video 5/6 ---

YouTube Link: <https://www.youtube.com/watch?v=dLbRF7oTl34>
CSV Path: video_csv/Video5.csv
Extracting cookies from chrome
Extracted 3444 cookies from chrome
[youtube] Extracting URL: <https://www.youtube.com/watch?v=dLbRF7oTl34>
[youtube] dLbRF7oTl34: Downloading webpage
[youtube] dLbRF7oTl34: Downloading tv client config
[youtube] dLbRF7oTl34: Downloading tv player API JSON
Extracting cookies from chrome
Extracted 3444 cookies from chrome
[youtube] Extracting URL: <https://www.youtube.com/watch?v=dLbRF7oTl34>
[youtube] dLbRF7oTl34: Downloading webpage
[youtube] dLbRF7oTl34: Downloading tv client config
[youtube] dLbRF7oTl34: Downloading tv player API JSON
[info] dLbRF7oTl34: Downloading 1 format(s): 18
[download] Destination: /Users/lmarte/Documents/Projects/CU-
Boulder/Deep_Learning/gridiron-ai/data/videos/2024 Las Vegas Raiders vs
Baltimore Ravens Week 2.mp4
[download] 100% of 365.76MiB in 00:00:15 at 24.18MiB/s
Downloaded video: /Users/lmarte/Documents/Projects/CU-
Boulder/Deep_Learning/gridiron-ai/data/videos/2024 Las Vegas Raiders vs
Baltimore Ravens Week 2.mp4
Loaded 11 timestamps from Video5
Saved: pass_0000_67.jpg
Saved: run_0001_162.jpg
Saved: run_0002_198.jpg
Saved: pass_0003_217.jpg
Saved: pass_0004_1006.jpg
Saved: pass_0005_1031.jpg

Saved: pass_0006_1083.jpg
Saved: run_0007_1118.jpg
Saved: run_0008_1337.jpg
Saved: run_0009_1362.jpg
Saved: run_0010_3400.jpg
Generated 55 screenshots
Deleted video: /Users/lmarte/Documents/Projects/CU-
Boulder/Deep_Learning/gridiron-ai/data/videos/2024 Las Vegas Raiders vs
Baltimore Ravens Week 2.mp4
Successfully processed Video5

--- Processing Video 6/6 ---

YouTube Link: <https://www.youtube.com/watch?v=fvaaMcpQBSE>
CSV Path: video_csv/Video6.csv
Extracting cookies from chrome
Extracted 3444 cookies from chrome
[youtube] Extracting URL: <https://www.youtube.com/watch?v=fvaaMcpQBSE>
[youtube] fvaaMcpQBSE: Downloading webpage
[youtube] fvaaMcpQBSE: Downloading tv client config
[youtube] fvaaMcpQBSE: Downloading tv player API JSON
Extracting cookies from chrome
Extracted 3444 cookies from chrome
[youtube] Extracting URL: <https://www.youtube.com/watch?v=fvaaMcpQBSE>
[youtube] fvaaMcpQBSE: Downloading webpage
[youtube] fvaaMcpQBSE: Downloading tv client config
[youtube] fvaaMcpQBSE: Downloading tv player API JSON
[info] fvaaMcpQBSE: Downloading 1 format(s): 18
[download] Destination: /Users/lmarte/Documents/Projects/CU-
Boulder/Deep_Learning/gridiron-ai/data/videos/2024 Baltimore Ravens vs
Washington Commanders Week 6.mp4
[download] 100% of 350.28MiB in 00:00:12 at 28.95MiB/s
Downloaded video: /Users/lmarte/Documents/Projects/CU-
Boulder/Deep_Learning/gridiron-ai/data/videos/2024 Baltimore Ravens vs
Washington Commanders Week 6.mp4
Loaded 11 timestamps from Video6
Saved: run_0000_109.jpg
Saved: pass_0001_134.jpg
Saved: run_0002_172.jpg
Saved: pass_0003_197.jpg
Saved: pass_0004_221.jpg
Saved: run_0005_255.jpg
Saved: pass_0006_1479.jpg
Saved: run_0007_2638.jpg
Saved: run_0008_2665.jpg
Saved: run_0009_2696.jpg
Saved: run_0010_2736.jpg
Generated 66 screenshots
Deleted video: /Users/lmarte/Documents/Projects/CU-

Boulder/Deep_Learning/gridiron-ai/data/videos/2024 Baltimore Ravens vs
Washington Commanders Week 6.mp4
Successfully processed Video6

Finished processing all 6 videos!

```
[2]: import sys
sys.path.append('../src')

import importlib
import feature_extraction
importlib.reload(feature_extraction)
from feature_extraction import FormationFeatureExtractor
import os
import json
import pandas as pd
from pathlib import Path
import numpy as np

# Initialize feature extractor
print("Initializing Feature Extractor...")
feature_extractor = FormationFeatureExtractor()

def extract_play_type_from_filename(filename, folder_name):
    """Extract play type from filename or use folder-based logic"""
    filename_lower = filename.lower()

    # Try to extract from filename first
    if filename_lower.startswith('run_') or 'run' in filename_lower:
        return 'run'
    elif filename_lower.startswith('pass_') or 'pass' in filename_lower:
        return 'pass'

    # If not clear from filename, return unknown for now
    return 'unknown'

def process_image_folder(folder_path, folder_name, dataset_split):
    """Process all images in a folder and extract features"""
    print(f"\nProcessing {folder_name} ({dataset_split}) folder...")

    if not folder_path.exists():
        print(f"Folder not found: {folder_path}")
        return [], []

    # Get all image files
    image_extensions = ['*.jpg', '*.jpeg', '*.png', '*.bmp']
    image_files = []
```

```

for ext in image_extensions:
    image_files.extend(folder_path.glob(ext))

print(f"    Found {len(image_files)} images")

features_list = []
failed_list = []

for i, image_path in enumerate(image_files):
    if i % 10 == 0: # Progress update every 10 images
        print(f"    Processing {i+1}/{len(image_files)}: {image_path.name}")

    try:
        # Extract features
        features = feature_extractor.extract_features(str(image_path))

        if features is not None:
            # Add metadata
            features['filename'] = image_path.name
            features['dataset_split'] = dataset_split
            features['source_folder'] = folder_name
            features['play_type'] = _
            ↪extract_play_type_from_filename(image_path.name, folder_name)

            features_list.append(features)
        else:
            failed_list.append(image_path.name)

    except Exception as e:
        print(f"Error processing {image_path.name}: {e}")
        failed_list.append(image_path.name)

print(f"Successfully processed: {len(features_list)} images")
print(f"Failed extractions: {len(failed_list)} images")

return features_list, failed_list

# =====
# PROCESS ALL IMAGE SOURCES
# =====

print("Comprehensive Feature Extraction")
print("=" * 50)

all_features = []
all_failed = []

```

```

# 1. Process YouTube Screenshots
print("\nYOUTUBE SCREENSHOTS")
screenshots_dir = Path("../data/youtube_screenshots")
youtube_features, youtube_failed = process_image_folder(screenshots_dir,
    ↪ "youtube_screenshots", "youtube")
all_features.extend(youtube_features)
all_failed.extend(youtube_failed)

# 2. Process Train/Valid/Test folders
data_folders = [
    ("../data/train/images", "train"),
    ("../data/valid/images", "valid"),
    ("../data/test/images", "test")
]

for folder_path_str, split_name in data_folders:
    folder_path = Path(folder_path_str)
    print(f"\n{len(all_features)//1000 + 2} {split_name.upper()} DATASET")

    split_features, split_failed = process_image_folder(folder_path,
    ↪ f"{split_name}_images", split_name)
    all_features.extend(split_features)
    all_failed.extend(split_failed)

# 3. Also check for alternative folder structures
alternative_folders = [
    ("../data/original/train/images", "train"),
    ("../data/original/valid/images", "valid"),
    ("../data/original/test/images", "test")
]

for folder_path_str, split_name in alternative_folders:
    folder_path = Path(folder_path_str)
    if folder_path.exists():
        print(f"\n ALTERNATIVE {split_name.upper()} FOLDER")
        split_features, split_failed = process_image_folder(folder_path,
    ↪ f"original_{split_name}", split_name)
        all_features.extend(split_features)
        all_failed.extend(split_failed)

# =====
# FINAL RESULTS AND SAVING
# =====

print(f"\nEXTRACTION COMPLETE!")
print("=" * 50)
print(f"Total images processed: {len(all_features)}")

```

```

print(f"Total failed extractions: {len(all_failed)}")

if all_features:
    # Convert to DataFrame
    features_df = pd.DataFrame(all_features)
    print(f"\nFeature DataFrame shape: {features_df.shape}")
    print(f"Total features per image: {len([col for col in features_df.columns_
    ↪if col not in ['filename', 'dataset_split', 'source_folder',
    ↪'play_type']])}")

    # Summary by source
    print(f"\nSummary by Source:")
    source_summary = features_df.groupby(['dataset_split', 'source_folder']).
    ↪size()
    for (split, source), count in source_summary.items():
        print(f"    {split:10s} - {source:20s}: {count:4d} images")

    # Summary by play type
    print(f"\nSummary by Play Type:")
    play_type_summary = features_df.groupby(['dataset_split', 'play_type']).
    ↪size()
    for (split, play_type), count in play_type_summary.items():
        print(f"    {split:10s} - {play_type:10s}: {count:4d} images")

    # Save comprehensive features
    features_csv_path = "../data/processed/comprehensive_features.csv"
    os.makedirs("../data/processed", exist_ok=True)
    features_df.to_csv(features_csv_path, index=False)
    print(f"\n Features saved to: {features_csv_path}")

    # Save separate files by dataset split
    for split in features_df['dataset_split'].unique():
        split_df = features_df[features_df['dataset_split'] == split]
        split_path = f"../data/processed/features_{split}.csv"
        split_df.to_csv(split_path, index=False)
        print(f"    {split} features saved to: {split_path}")

else:
    print("No features extracted successfully!")

if all_failed:
    print(f"\nFailed files sample: {all_failed[:10]}")

    # Save failed files list
    failed_path = "../data/processed/failed_extractions.txt"
    with open(failed_path, 'w') as f:
        for failed_file in all_failed:

```

```
f.write(f"{failed_file}\n")
print(f"Failed files list saved to: {failed_path}")
```

Initializing Feature Extractor...
Comprehensive Feature Extraction

=====

YOUTUBE SCREENSHOTS

Processing youtube_screenshots (youtube) folder...

Found 66 images

Processing 1/66: pass_0009_3295.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0009_3295.jpg: 384x640 18 defenses, 3 olines, 5 wide_receivers, 101.1ms
Speed: 3.1ms preprocess, 101.1ms inference, 4.5ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0000_67.jpg: 384x640 13 defenses, 8 olines, 1 ref, 2 running_backs, 6 wide_receivers, 73.2ms
Speed: 0.4ms preprocess, 73.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0008_1443.jpg: 384x640 16 defenses, 5 olines, 1 ref, 3 wide_receivers, 145.6ms
Speed: 0.5ms preprocess, 145.6ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0007_2632.jpg: 384x640 17 defenses, 3 olines, 2 running_backs, 5 wide_receivers, 75.9ms
Speed: 0.7ms preprocess, 75.9ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0000_109.jpg: 384x640 15 defenses, 1 oline, 1 ref, 7 wide_receivers, 71.1ms
Speed: 0.5ms preprocess, 71.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0007_1311.jpg: 384x640 14 defenses, 4 olines, 1 qb, 2 refs, 1 running_back, 10 wide_receivers, 72.0ms
Speed: 0.5ms preprocess, 72.0ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0002_118.jpg: 384x640 15 defenses, 5 olines, 1 ref, 1 running_back, 8 wide_receivers, 72.6ms
Speed: 0.5ms preprocess, 72.6ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0008_1337.jpg: 384x640 13 defenses, 6 olines, 5 refs, 2 running_backs, 3 wide_receivers, 75.0ms
Speed: 0.6ms preprocess, 75.0ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0005_1031.jpg: 384x640 19 defenses, 7 olines, 1 running_back, 2 wide_receivers, 73.2ms
Speed: 0.5ms preprocess, 73.2ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0010_3670.jpg: 384x640 17 defenses, 8 olines, 1 ref, 2 running_backs, 5 wide_receivers, 72.9ms
Speed: 0.5ms preprocess, 72.9ms inference, 0.7ms postprocess per image at shape (1, 3, 384, 640)

Processing 11/66: run_0000_127.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0000_127.jpg: 384x640 17 defenses, 4 olines, 4 wide_receivers, 72.2ms
Speed: 0.4ms preprocess, 72.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0003_146.jpg: 384x640 13 defenses, 6 olines, 1 ref, 3 wide_receivers, 71.8ms
Speed: 0.5ms preprocess, 71.8ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0010_3199.jpg: 384x640 13 defenses, 8 olines, 4 refs, 4 wide_receivers, 72.0ms
Speed: 0.5ms preprocess, 72.0ms inference, 0.7ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0003_197.jpg: 384x640 18 defenses, 6 olines, 3 wide_receivers, 88.4ms
Speed: 0.6ms preprocess, 88.4ms inference, 0.7ms postprocess per image at shape

(1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0006_777.jpg: 384x640 15 defenses, 7 olines, 7 wide_receivers, 74.0ms
Speed: 0.5ms preprocess, 74.0ms inference, 0.9ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0001_51.jpg: 384x640 15 defenses, 7 olines, 4 wide_receivers, 75.3ms
Speed: 0.5ms preprocess, 75.3ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0002_540.jpg: 384x640 17 defenses, 2 olines, 1 ref, 5 wide_receivers, 70.4ms
Speed: 0.5ms preprocess, 70.4ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0008_2665.jpg: 384x640 15 defenses, 4 olines, 2 qbs, 2 refs, 2 running_backs, 5 wide_receivers, 71.2ms
Speed: 0.5ms preprocess, 71.2ms inference, 0.8ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0005_1281.jpg: 384x640 17 defenses, 4 olines, 5 wide_receivers, 73.6ms
Speed: 0.5ms preprocess, 73.6ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0007_2479.jpg: 384x640 17 defenses, 1 online, 3 refs, 4 wide_receivers, 71.4ms
Speed: 0.5ms preprocess, 71.4ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

Processing 21/66: run_0010_2736.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0010_2736.jpg: 384x640 14 defenses, 3 olines, 4 refs, 2 running_backs, 4 wide_receivers, 79.5ms
Speed: 0.5ms preprocess, 79.5ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0010_3400.jpg: 384x640 15 defenses, 2 olines, 1 qb, 9 refs, 2 running_backs, 2 wide_receivers, 73.8ms

Speed: 0.5ms preprocess, 73.8ms inference, 0.8ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0008_2672.jpg: 384x640 16 defenses, 2 olines, 1 qb, 2 running_backs, 11 wide_receivers, 74.2ms
Speed: 0.4ms preprocess, 74.2ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0000_30.jpg: 384x640 14 defenses, 9 olines, 1 qb, 3 running_backs, 5 wide_receivers, 71.8ms
Speed: 0.5ms preprocess, 71.8ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0005_1253.jpg: 384x640 15 defenses, 1 oline, 1 ref, 5 wide_receivers, 71.9ms
Speed: 0.5ms preprocess, 71.9ms inference, 0.8ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0004_1052.jpg: 384x640 17 defenses, 6 olines, 5 wide_receivers, 69.7ms
Speed: 0.4ms preprocess, 69.7ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0006_1389.jpg: 384x640 13 defenses, 8 olines, 4 wide_receivers, 69.8ms
Speed: 0.4ms preprocess, 69.8ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0004_887.jpg: 384x640 18 defenses, 2 olines, 4 wide_receivers, 74.0ms
Speed: 0.4ms preprocess, 74.0ms inference, 1.3ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0010_3996.jpg: 384x640 17 defenses, 3 olines, 1 ref, 7 wide_receivers, 76.5ms
Speed: 0.4ms preprocess, 76.5ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0001_134.jpg: 384x640 15 defenses, 2 olines, 4 refs, 7 wide_receivers, 72.1ms

Speed: 0.5ms preprocess, 72.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 640)

Processing 31/66: run_0003_695.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0003_695.jpg: 384x640 14 defenses, 9 olines, 1 running_back, 4 wide_receivers, 72.3ms
Speed: 0.5ms preprocess, 72.3ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0007_1413.jpg: 384x640 13 defenses, 7 olines, 1 ref, 7 wide_receivers, 70.2ms
Speed: 0.4ms preprocess, 70.2ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0001_55.jpg: 384x640 15 defenses, 4 olines, 5 wide_receivers, 69.3ms
Speed: 0.5ms preprocess, 69.3ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0009_2901.jpg: 384x640 15 defenses, 4 olines, 3 wide_receivers, 73.2ms
Speed: 0.5ms preprocess, 73.2ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0004_653.jpg: 384x640 12 defenses, 8 olines, 1 ref, 1 running_back, 3 wide_receivers, 70.5ms
Speed: 0.4ms preprocess, 70.5ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0003_217.jpg: 384x640 12 defenses, 10 olines, 2 refs, 2 running_backs, 2 wide_receivers, 73.0ms
Speed: 0.5ms preprocess, 73.0ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0003_564.jpg: 384x640 15 defenses, 3 olines, 1 ref, 1 running_back, 6 wide_receivers, 72.5ms
Speed: 0.5ms preprocess, 72.5ms inference, 0.7ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0000_25.jpg: 384x640 14 defenses,

9 olines, 5 wide_receivers, 69.5ms
Speed: 0.4ms preprocess, 69.5ms inference, 0.5ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-
ai/notebooks/./data/youtube_screenshots/pass_0008_2518.jpg: 384x640 17
defenses, 1 oline, 4 wide_receivers, 70.1ms
Speed: 0.4ms preprocess, 70.1ms inference, 0.6ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-
ai/notebooks/./data/youtube_screenshots/pass_0000_26.jpg: 384x640 19 defenses,
5 olines, 1 running_back, 7 wide_receivers, 71.5ms
Speed: 0.6ms preprocess, 71.5ms inference, 0.7ms postprocess per image at shape
(1, 3, 384, 640)

Processing 41/66: run_0009_2696.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-
ai/notebooks/./data/youtube_screenshots/run_0009_2696.jpg: 384x640 16 defenses,
7 olines, 1 ref, 1 running_back, 1 wide_receiver, 71.1ms
Speed: 0.4ms preprocess, 71.1ms inference, 0.7ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-
ai/notebooks/./data/youtube_screenshots/pass_0001_88.jpg: 384x640 18 defenses,
5 olines, 2 refs, 1 running_back, 8 wide_receivers, 70.0ms
Speed: 0.4ms preprocess, 70.0ms inference, 0.7ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-
ai/notebooks/./data/youtube_screenshots/run_0001_360.jpg: 384x640 12 defenses,
6 olines, 2 refs, 5 wide_receivers, 70.1ms
Speed: 0.4ms preprocess, 70.1ms inference, 0.5ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-
ai/notebooks/./data/youtube_screenshots/run_0001_162.jpg: 384x640 13 defenses,
10 olines, 6 refs, 3 wide_receivers, 70.4ms
Speed: 0.4ms preprocess, 70.4ms inference, 0.4ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-
ai/notebooks/./data/youtube_screenshots/run_0005_752.jpg: 384x640 18 defenses,
7 olines, 1 ref, 70.2ms
Speed: 0.5ms preprocess, 70.2ms inference, 0.6ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-

ai/notebooks/./data/youtube_screenshots/run_0009_1362.jpg: 384x640 15 defenses, 4 olines, 1 qb, 4 refs, 2 running_backs, 1 tight_end, 4 wide_receivers, 69.5ms
Speed: 0.4ms preprocess, 69.5ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0007_2638.jpg: 384x640 14 defenses, 6 olines, 1 qb, 2 refs, 2 running_backs, 4 wide_receivers, 73.3ms
Speed: 0.5ms preprocess, 73.3ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0006_1722.jpg: 384x640 16 defenses, 1 oline, 6 wide_receivers, 71.1ms
Speed: 0.5ms preprocess, 71.1ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0005_683.jpg: 384x640 14 defenses, 10 olines, 1 ref, 1 wide_receiver, 71.3ms
Speed: 0.7ms preprocess, 71.3ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0002_95.jpg: 384x640 17 defenses, 6 olines, 7 wide_receivers, 69.7ms
Speed: 0.5ms preprocess, 69.7ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

Processing 51/66: run_0002_172.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0002_172.jpg: 384x640 15 defenses, 5 olines, 1 running_back, 3 wide_receivers, 104.7ms
Speed: 0.4ms preprocess, 104.7ms inference, 1.0ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0002_198.jpg: 384x640 15 defenses, 7 olines, 1 running_back, 3 wide_receivers, 78.8ms
Speed: 0.5ms preprocess, 78.8ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0004_221.jpg: 384x640 13 defenses, 4 olines, 1 ref, 2 wide_receivers, 71.4ms
Speed: 0.5ms preprocess, 71.4ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0002_106.jpg: 384x640 11 defenses, 7 olines, 5 wide_receivers, 72.1ms
Speed: 0.5ms preprocess, 72.1ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0009_1884.jpg: 384x640 15 defenses, 5 olines, 2 refs, 1 running_back, 4 wide_receivers, 72.0ms
Speed: 0.5ms preprocess, 72.0ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0007_1118.jpg: 384x640 17 defenses, 5 olines, 1 qb, 4 refs, 1 running_back, 5 wide_receivers, 74.4ms
Speed: 0.5ms preprocess, 74.4ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0010_2822.jpg: 384x640 15 defenses, 4 olines, 3 wide_receivers, 72.8ms
Speed: 0.4ms preprocess, 72.8ms inference, 0.8ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0004_722.jpg: 384x640 15 defenses, 7 olines, 1 ref, 1 running_back, 1 tight_end, 6 wide_receivers, 71.3ms
Speed: 0.5ms preprocess, 71.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0008_1857.jpg: 384x640 16 defenses, 4 olines, 5 wide_receivers, 71.8ms
Speed: 0.5ms preprocess, 71.8ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/run_0005_255.jpg: 384x640 16 defenses, 3 olines, 1 qb, 2 refs, 4 running_backs, 3 wide_receivers, 70.6ms
Speed: 0.4ms preprocess, 70.6ms inference, 0.7ms postprocess per image at shape (1, 3, 384, 640)

Processing 61/66: pass_0004_1006.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/youtube_screenshots/pass_0004_1006.jpg: 384x640 15 defenses, 7 olines, 2 refs, 1 running_back, 2 wide_receivers, 71.3ms
Speed: 0.5ms preprocess, 71.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/youtube_screenshots/pass_0009_2794.jpg: 384x640 16 defenses, 5 olines, 6 wide_receivers, 71.1ms
Speed: 0.5ms preprocess, 71.1ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/youtube_screenshots/pass_0003_503.jpg: 384x640 17 defenses, 3 olines, 2 refs, 5 wide_receivers, 72.3ms
Speed: 0.5ms preprocess, 72.3ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/youtube_screenshots/pass_0006_1479.jpg: 384x640 13 defenses, 6 olines, 1 qb, 1 ref, 4 running_backs, 3 wide_receivers, 69.2ms
Speed: 0.4ms preprocess, 69.2ms inference, 1.0ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/youtube_screenshots/run_0006_1282.jpg: 384x640 16 defenses, 4 olines, 1 ref, 1 running_back, 4 wide_receivers, 72.2ms
Speed: 0.5ms preprocess, 72.2ms inference, 0.7ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/youtube_screenshots/pass_0006_1083.jpg: 384x640 15 defenses, 9 olines, 1 ref, 1 running_back, 2 wide_receivers, 70.9ms
Speed: 0.5ms preprocess, 70.9ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)
Successfully processed: 66 images
Failed extractions: 0 images

2 TRAIN DATASET

Processing train_images (train) folder...
Folder not found: ../data/train/images

2 VALID DATASET

Processing valid_images (valid) folder...
Folder not found: ../data/valid/images

2 TEST DATASET

Processing test_images (test) folder...
Folder not found: ../data/test/images

ALTERNATIVE TRAIN FOLDER

Processing original_train (train) folder...

Found 200 images

Processing 1/200:

58073_003389_Sideline_frame51_jpg.rf.2cfb4642415625f23cf051632716a23d.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/58073_003389_Sideline_frame51_jpg.rf.2cfb4642415625f23cf051632716a23d.jpg: 640x640 22 defenses, 2 olines, 2 refs, 1 running_back, 3 wide_receivers, 107.5ms
Speed: 0.6ms preprocess, 107.5ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture_PNG_jpg.rf.4a17b15578b25bf93736a fae608616e6.jpg: 640x640 13 defenses, 9 olines, 2 qbs, 1 ref, 1 running_back, 7 wide_receivers, 108.5ms
Speed: 0.6ms preprocess, 108.5ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture78_PNG_jpg.rf.9c4000aec94e1416314 94b7fda42c510.jpg: 640x640 15 defenses, 6 olines, 2 qbs, 1 ref, 2 running_backs, 8 wide_receivers, 104.2ms
Speed: 0.7ms preprocess, 104.2ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture179_PNG_jpg.rf.117be4fe0d2aedbd88 abd062ea20c68c.jpg: 640x640 16 defenses, 4 olines, 2 qbs, 3 refs, 4 running_backs, 6 wide_receivers, 102.5ms
Speed: 0.7ms preprocess, 102.5ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture121_PNG_jpg.rf.3d57a4095fec2c2020 0d442e6992cb1c.jpg: 640x640 17 defenses, 6 olines, 3 wide_receivers, 105.4ms
Speed: 0.9ms preprocess, 105.4ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture11_PNG_jpg.rf.f492dd98a1aa24f4309 b46c646353641.jpg: 640x640 17 defenses, 6 olines, 1 qb, 4 refs, 2 running_backs, 3 wide_receivers, 102.4ms
Speed: 0.6ms preprocess, 102.4ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57890_000485_Sideline_frame93_jpg.rf.5304e6b3337e541a865273464e924303.jpg: 640x640 12 defenses, 7 olines, 2 qbs, 5 refs, 3 running_backs, 6 wide_receivers, 101.4ms
Speed: 0.7ms preprocess, 101.4ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture72_PNG_jpg.rf.4614a8ad4a918c4ce19c642784605d2b.jpg: 640x640 13 defenses, 6 qbs, 2 refs, 4 running_backs, 4 wide_receivers, 109.3ms
Speed: 0.6ms preprocess, 109.3ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture16_PNG_jpg.rf.d82481f7cf8ca1be3e6266fdec05d493.jpg: 640x640 13 defenses, 6 olines, 2 qbs, 1 ref, 4 running_backs, 3 wide_receivers, 108.8ms
Speed: 0.7ms preprocess, 108.8ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57594_004453_Sideline_frame102_jpg.rf.5297d541b4604ffde617703e8c38fd4d.jpg: 640x640 15 defenses, 8 olines, 1 qb, 2 running_backs, 6 wide_receivers, 101.3ms
Speed: 0.8ms preprocess, 101.3ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

Processing 11/200:

Oregon_0_vs_Colorado_2019_mp4-85_jpg.rf.902135149a6b109b084f8ab59a5bf8b7.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Oregon_0_vs_Colorado_2019_mp4-85_jpg.rf.902135149a6b109b084f8ab59a5bf8b7.jpg: 640x640 17 defenses, 5 olines, 4 refs, 1 running_back, 6 wide_receivers, 107.0ms
Speed: 1.6ms preprocess, 107.0ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57940_003225_Sideline_frame215_jpg.rf.2ecfe0e76db00e67d60f031115d07398.jpg: 640x640 15 defenses, 2 olines, 3 qbs, 2 refs, 5 running_backs, 5 wide_receivers, 101.7ms
Speed: 0.6ms preprocess, 101.7ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-154_jpg.rf.eda53bbc7a070701d1ff1f0937332408.jpg: 640x640 12 defenses, 6 olines, 3 refs, 3 running_backs, 4 wide_receivers,

102.3ms

Speed: 0.6ms preprocess, 102.3ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture7_PNG_jpg.rf.cfb16626da387f3c77ca a278681033d9.jpg: 640x640 14 defenses, 6 olines, 2 qbs, 1 running_back, 4 wide_receivers, 104.8ms

Speed: 0.7ms preprocess, 104.8ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Sideline4_mp4-0_jpg.rf.662c0536cb521df050e61dc041cb88af.jpg: 640x640 17 defenses, 1 oline, 6 refs, 1 running_back, 6 wide_receivers, 99.9ms

Speed: 0.6ms preprocess, 99.9ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57733_001017_Sideline_frame71_jpg.rf.f5b c489f3ec1afcc9c6b75d5403bb661.jpg: 640x640 19 defenses, 1 oline, 2 qbs, 8 refs, 3 running_backs, 4 wide_receivers, 102.3ms

Speed: 0.6ms preprocess, 102.3ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57895_000807_Sideline_frame34_jpg.rf.142 7ddae4e9f27ef4f9925c896cfc6ee.jpg: 640x640 17 defenses, 4 olines, 8 qbs, 1 ref, 4 running_backs, 2 wide_receivers, 102.0ms

Speed: 0.6ms preprocess, 102.0ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture58_PNG_jpg.rf.6381df0a329035a3698 61d819c5cd219.jpg: 640x640 14 defenses, 5 olines, 2 qbs, 2 refs, 1 running_back, 6 wide_receivers, 102.2ms

Speed: 0.7ms preprocess, 102.2ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-76_jpg.rf.6a33f5d6a0cb8a578ba9bc1838620874.jpg: 640x640 13 defenses, 9 olines, 3 qbs, 1 ref, 2 running_backs, 1 tight_end, 6 wide_receivers, 101.5ms

Speed: 0.7ms preprocess, 101.5ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/

notebooks/../../data/original/train/images/Capture_PNG_jpg.rf.a68d41425d7d96b7ca39c60badbb6108.jpg: 640x640 12 defenses, 12 olines, 1 ref, 3 wide_receivers, 99.6ms
Speed: 0.6ms preprocess, 99.6ms inference, 0.5ms postprocess per image at shape
(1, 3, 640, 640)

Processing 21/200: Capture20_PNG_jpg.rf.d6769ce6bca562a7d7cff8bd9f8d98df.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/Capture20_PNG_jpg.rf.d6769ce6bca562a7d7c
ff8bd9f8d98df.jpg: 640x640 16 defenses, 4 olines, 4 qbs, 3 refs, 4
running_backs, 2 wide_receivers, 148.6ms
Speed: 0.6ms preprocess, 148.6ms inference, 1.5ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/Capture17_PNG_jpg.rf.f7414500f4690197cd1
cc9d3542c458e.jpg: 640x640 18 defenses, 3 olines, 2 qbs, 6 refs, 2
running_backs, 3 wide_receivers, 104.2ms
Speed: 0.8ms preprocess, 104.2ms inference, 0.6ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/Oregon_0_vs_Colorado_2019_mp4-
53_jpg.rf.645004f972f7301a33c2571785d91a23.jpg: 640x640 13 defenses, 8 olines, 2
qbs, 3 refs, 2 running_backs, 4 wide_receivers, 107.5ms
Speed: 0.7ms preprocess, 107.5ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/58095_002347_Sideline_frame107_jpg.rf.71
620fcc5c31b37f070477059ba04570.jpg: 640x640 17 defenses, 5 olines, 1 qb, 13
refs, 2 running_backs, 3 wide_receivers, 103.6ms
Speed: 0.6ms preprocess, 103.6ms inference, 0.5ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-
ai/notebooks/../../data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-
Defense-All-22-Cut-up-480p-mp4-130_jpg.rf.9c96cc3d8b5712c8c705bc2913b80f1b.jpg:
640x640 14 defenses, 4 olines, 2 qbs, 4 wide_receivers, 115.6ms
Speed: 0.8ms preprocess, 115.6ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/cap-4-
_jpg.rf.c0c99c77800e77d349b90276e5dadd82.jpg: 640x640 13 defenses, 7 olines, 2
refs, 2 running_backs, 6 wide_receivers, 111.6ms
Speed: 0.7ms preprocess, 111.6ms inference, 0.5ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57700_002907_Sideline_frame118_jpg.rf.499a88dcc5b66e52b68158b219a98e76.jpg: 640x640 11 defenses, 9 olines, 3 qbs, 3 running_backs, 3 wide_receivers, 102.2ms
Speed: 0.6ms preprocess, 102.2ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture224_PNG_jpg.rf.a44ead9cd0b28fa93fe271be4ccd4e9c.jpg: 640x640 14 defenses, 4 olines, 5 qbs, 2 running_backs, 4 wide_receivers, 138.1ms
Speed: 0.6ms preprocess, 138.1ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57581_002177_Sideline_frame181_jpg.rf.e0e3a89fc07fa8ce51fb5af8f58be055.jpg: 640x640 16 defenses, 10 olines, 1 ref, 2 wide_receivers, 101.7ms
Speed: 0.6ms preprocess, 101.7ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture21_PNG_jpg.rf.a1f44f161eb39980ee1020339a30e311.jpg: 640x640 16 defenses, 5 olines, 2 qbs, 2 running_backs, 6 wide_receivers, 102.0ms
Speed: 0.6ms preprocess, 102.0ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

Processing 31/200:

58157_001501_Sideline_frame140_jpg.rf.01841ee3655682a4d2569b72a3581a5c.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/58157_001501_Sideline_frame140_jpg.rf.01841ee3655682a4d2569b72a3581a5c.jpg: 640x640 21 defenses, 2 olines, 4 refs, 4 wide_receivers, 104.8ms
Speed: 0.6ms preprocess, 104.8ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57619_001198_Sideline_frame144_jpg.rf.fdba37b841454f1f085a97dcc52b828c.jpg: 640x640 14 defenses, 5 olines, 1 running_back, 5 wide_receivers, 109.9ms
Speed: 0.8ms preprocess, 109.9ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/58149_001082_Sideline_frame73_jpg.rf.8b057ee478052829a8a817b9da400113.jpg: 640x640 19 defenses, 3 olines, 1 qb, 2 refs, 1 running_back, 6 wide_receivers, 105.5ms

Speed: 0.7ms preprocess, 105.5ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture14_PNG_jpg.rf.44ee1f0681a3d7864be387bef1feec8d.jpg: 640x640 15 defenses, 4 olines, 1 ref, 4 wide_receivers, 102.2ms

Speed: 0.7ms preprocess, 102.2ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/58080_000869_Sideline_frame81_jpg.rf.3c67f692c921a825d36229727bf00d34.jpg: 640x640 21 defenses, 3 olines, 4 refs, 8 wide_receivers, 101.8ms

Speed: 0.8ms preprocess, 101.8ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture124_PNG_jpg.rf.ded1e16850c3382b0a9cc07b2b249d14.jpg: 640x640 15 defenses, 9 olines, 2 refs, 3 wide_receivers, 101.8ms

Speed: 0.7ms preprocess, 101.8ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture130_PNG_jpg.rf.637f157c40297e3455bc267acf68bcc8.jpg: 640x640 12 defenses, 4 olines, 1 qb, 1 ref, 3 running_backs, 2 wide_receivers, 175.2ms

Speed: 0.6ms preprocess, 175.2ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57516_001350_Sideline_frame140_jpg.rf.8823e0c3295ebfc1c6df922ab2525a11.jpg: 640x640 20 defenses, 1 oline, 5 qbs, 6 refs, 5 running_backs, 4 wide_receivers, 107.2ms

Speed: 0.7ms preprocess, 107.2ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57526_001984_Sideline_frame50_jpg.rf.8e79229d2205a04baf7cf24d3e934769.jpg: 640x640 15 defenses, 2 olines, 5 qbs, 2 running_backs, 9 wide_receivers, 100.9ms

Speed: 0.7ms preprocess, 100.9ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57756_002564_Sideline_frame30_jpg.rf.ce8c8e733e8ef1647b26f5e5b5e833ea.jpg: 640x640 16 defenses, 1 oline, 5 refs, 1

running_back, 5 wide_receivers, 107.8ms
Speed: 0.6ms preprocess, 107.8ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

Processing 41/200: Capture128_PNG_jpg.rf.cf302039ab5f9c75caa300dddd07b94d.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture128_PNG_jpg.rf.cf302039ab5f9c75caa300dddd07b94d.jpg: 640x640 14 defenses, 5 olines, 2 qbs, 3 running_backs, 7 wide_receivers, 103.7ms
Speed: 0.6ms preprocess, 103.7ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture125_PNG_jpg.rf.bec0dd600d1f774a7c41fff228bbe3a7.jpg: 640x640 16 defenses, 3 olines, 5 qbs, 4 refs, 6 running_backs, 4 wide_receivers, 101.1ms
Speed: 0.6ms preprocess, 101.1ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture202_PNG_jpg.rf.c904d2c7b42020fcb52ebc819244cc8c.jpg: 640x640 14 defenses, 7 olines, 2 qbs, 2 refs, 2 running_backs, 5 wide_receivers, 101.1ms
Speed: 0.6ms preprocess, 101.1ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture21_PNG_jpg.rf.f48876f1949e50aedf41e3eafaec6088.jpg: 640x640 12 defenses, 10 olines, 3 qbs, 1 ref, 2 running_backs, 3 wide_receivers, 105.8ms
Speed: 0.6ms preprocess, 105.8ms inference, 1.4ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-mp4-199_jpg.rf.ed3372e7c914eae71eaf74e8d42f8788.jpg: 640x640 13 defenses, 6 olines, 3 qbs, 3 refs, 5 running_backs, 5 wide_receivers, 106.1ms
Speed: 0.8ms preprocess, 106.1ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Bengals_mp4-2_jpg.rf.70e1fa74f87bdb9d2bda358632f7c25e.jpg: 640x640 12 defenses, 4 olines, 1 ref, 3 running_backs, 6 wide_receivers, 102.5ms
Speed: 0.7ms preprocess, 102.5ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/videoplayback_mp4-5_jpg.rf.0389a4020ce95b858b6fd65cfa170939.jpg: 640x640 15 defenses, 5 olines, 5 qbs, 5 refs, 1 running_back, 3 wide_receivers, 109.7ms
Speed: 0.6ms preprocess, 109.7ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Bengals_mp4-2_jpg.rf.babee70c59884d388f756121f94a7c92.jpg: 640x640 12 defenses, 4 olines, 2 refs, 2 running_backs, 5 wide_receivers, 101.9ms
Speed: 0.6ms preprocess, 101.9ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture94_PNG_jpg.rf.8f4f1c18ffaa706dc83fe2f308e020b3.jpg: 640x640 15 defenses, 5 olines, 5 wide_receivers, 102.2ms
Speed: 0.7ms preprocess, 102.2ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture33_PNG_jpg.rf.1807d2820bf196729c9908dd14c86867.jpg: 640x640 18 defenses, 8 olines, 3 refs, 1 running_back, 2 wide_receivers, 101.3ms
Speed: 0.6ms preprocess, 101.3ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

Processing 51/200:

57833_001229_Sideline_frame0787_jpg.rf.122faf6ffe9b239d906c48ba3d00f701.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57833_001229_Sideline_frame0787_jpg.rf.122faf6ffe9b239d906c48ba3d00f701.jpg: 640x640 14 defenses, 5 olines, 1 qb, 1 ref, 2 running_backs, 6 wide_receivers, 103.0ms
Speed: 0.6ms preprocess, 103.0ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture19_PNG_jpg.rf.637491fe5455010df69d70739ac7a13b.jpg: 640x640 10 defenses, 4 olines, 1 qb, 4 refs, 1 running_back, 4 wide_receivers, 116.9ms
Speed: 0.7ms preprocess, 116.9ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57509_000467_Sideline_frame147_jpg.rf.e9b8e2600fdfedea46ec4a73ef926d17.jpg: 640x640 12 defenses, 2 olines, 3 qbs, 2 refs, 1 running_back, 5 wide_receivers, 113.5ms
Speed: 0.8ms preprocess, 113.5ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58152_002751_Sideline_frame171_jpg.rf.616d3640e3cd88f2d4def18f2ecb5351.jpg: 640x640 14 defenses, 4 olines, 4 qbs, 3 running_backs, 5 wide_receivers, 102.2ms
Speed: 0.6ms preprocess, 102.2ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture140_PNG_jpg.rf.980e2b787e9ea41d9550b3a9fe6d863b.jpg: 640x640 18 defenses, 11 olines, 1 ref, 4 wide_receivers, 99.3ms
Speed: 0.6ms preprocess, 99.3ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture13_PNG_jpg.rf.8c77946f70d49900f96ee6373f107c.jpg: 640x640 16 defenses, 3 olines, 9 refs, 14 wide_receivers, 105.5ms
Speed: 0.6ms preprocess, 105.5ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/field_png.rf.1c03877c6fb9cb3848d986fd3df4cc3a.jpg: 640x640 23 defenses, 6 olines, 1 qb, 1 ref, 1 running_back, 7 wide_receivers, 98.3ms
Speed: 0.6ms preprocess, 98.3ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57578_002753_Sideline_frame0829_jpg.rf.1212d4f87c55fde95218c38bb779a8c2.jpg: 640x640 17 defenses, 1 qb, 1 ref, 1 running_back, 10 wide_receivers, 116.6ms
Speed: 0.6ms preprocess, 116.6ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58144_003352_Sideline_frame0632_jpg.rf.d2ddc1eaed77feb354f07336752143d.jpg: 640x640 19 defenses, 1 oline, 2 refs, 1 running_back, 2 wide_receivers, 105.3ms
Speed: 0.6ms preprocess, 105.3ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57540_000276_Sideline_frame37_jpg.rf.076be97405de4ba360acf7dedc932341.jpg: 640x640 12 defenses, 12 olines, 6 refs, 1 running_back, 5 wide_receivers, 100.4ms

Speed: 0.6ms preprocess, 100.4ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

Processing 61/200:

57629_003086_Sideline_frame0540_jpg.rf.3a30c54a080e3f5bcb92f327e06ab69e.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57629_003086_Sideline_frame0540_jpg.rf.3a30c54a080e3f5bcb92f327e06ab69e.jpg: 640x640 12 defenses, 10 olines, 3 qbs, 17 refs, 3 running_backs, 3 wide_receivers, 99.1ms

Speed: 0.7ms preprocess, 99.1ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57568_003210_Sideline_frame106_jpg.rf.7a58eb91fc117c02ece5eac19cef92a0.jpg: 640x640 12 defenses, 3 olines, 3 qbs, 5 refs, 2 running_backs, 8 wide_receivers, 99.1ms

Speed: 0.7ms preprocess, 99.1ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture185_PNG_jpg.rf.6fdf9da696a6bb9d05ae5a839069413c.jpg: 640x640 17 defenses, 6 olines, 2 refs, 6 wide_receivers, 109.8ms

Speed: 0.7ms preprocess, 109.8ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57988_001226_Sideline_frame70_jpg.rf.7d48b0c7920d6631093130243e70c985.jpg: 640x640 15 defenses, 5 olines, 2 qbs, 5 refs, 1 running_back, 7 wide_receivers, 102.3ms

Speed: 0.6ms preprocess, 102.3ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture150_PNG_jpg.rf.653f942edb541f57659100c32694ebc0.jpg: 640x640 13 defenses, 3 olines, 4 qbs, 1 ref, 3 running_backs, 4 wide_receivers, 100.9ms

Speed: 0.6ms preprocess, 100.9ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture58_PNG_jpg.rf.39cad65af941a653ded20a0f628ddccf.jpg: 640x640 12 defenses, 8 olines, 3 qbs, 3 refs, 3 running_backs, 7 wide_receivers, 102.1ms

Speed: 0.8ms preprocess, 102.1ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/

notebooks/../../data/original/train/images/57943_003025_Sideline_frame1000_jpg.rf.d
e265cea6294c2252ea929bf9418f16e.jpg: 640x640 14 defenses, 1 oline, 7 qbs, 5
refs, 2 running_backs, 5 wide_receivers, 140.0ms
Speed: 0.6ms preprocess, 140.0ms inference, 1.0ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/57519_000109_Sideline_frame217_jpg.rf.05
cf60fe8926c86d1a4147dd9f98d41c.jpg: 640x640 16 defenses, 7 olines, 1 ref, 1
running_back, 3 wide_receivers, 100.9ms
Speed: 0.7ms preprocess, 100.9ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/57930_004167_Sideline_frame111_jpg.rf.8a
69c3a6f9f1c8bcc8521196121fec21.jpg: 640x640 18 defenses, 4 olines, 3 qbs, 3
refs, 2 running_backs, 6 wide_receivers, 98.7ms
Speed: 0.7ms preprocess, 98.7ms inference, 0.9ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/videoplayback_mp4-
27_jpg.rf.5398332ebd06370a7cb53c27dcf802f.jpg: 640x640 16 defenses, 10 olines,
2 qbs, 6 refs, 2 running_backs, 3 wide_receivers, 100.0ms
Speed: 1.0ms preprocess, 100.0ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 640)

Processing 71/200: Capture85_PNG_jpg.rf.28e895fd97191cb703af488a690039cc.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/Capture85_PNG_jpg.rf.28e895fd97191cb703a
f488a690039cc.jpg: 640x640 11 defenses, 6 olines, 9 qbs, 5 running_backs, 4
wide_receivers, 99.7ms
Speed: 0.6ms preprocess, 99.7ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/Capture147_PNG_jpg.rf.950c0f73ed15de3c4c
2fb99099efebc4.jpg: 640x640 16 defenses, 9 olines, 1 qb, 7 refs, 2
running_backs, 5 wide_receivers, 100.3ms
Speed: 0.6ms preprocess, 100.3ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/WashSt_vs_IowaSt2018_mp4-
51_jpg.rf.09d1cdd1cdab80cdd2b61ef26c91a53a.jpg: 640x640 20 defenses, 1 oline, 9
refs, 1 running_back, 7 wide_receivers, 100.2ms
Speed: 0.6ms preprocess, 100.2ms inference, 1.0ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Screen-Recording-2023-09-20-at-6_54_05-PM_mov-0_jpg.rf.7eae047ce52281b3e29f55824ccf97e7.jpg: 640x640 19 defenses, 5 olines, 1 qb, 5 wide_receivers, 101.3ms
Speed: 0.6ms preprocess, 101.3ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Sideline2_mp4-0_jpg.rf.79520c214748c502a152a16be57f7623.jpg: 640x640 18 defenses, 3 olines, 3 refs, 1 running_back, 5 wide_receivers, 102.8ms
Speed: 0.7ms preprocess, 102.8ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture_PNG_jpg.rf.db2ed5fbb7b3f493bc45be94b506a979.jpg: 640x640 17 defenses, 6 olines, 2 wide_receivers, 102.9ms
Speed: 0.6ms preprocess, 102.9ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57575_003430_Sideline_frame146_jpg.rf.8eefa569452b31ca0376ac7d4606fd7e.jpg: 640x640 22 defenses, 4 olines, 3 running_backs, 7 wide_receivers, 103.6ms
Speed: 0.7ms preprocess, 103.6ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/58013_003447_Sideline_frame0640_jpg.rf.e9ce07369bf1088055d4b56cce20a588.jpg: 640x640 15 defenses, 3 olines, 1 ref, 1 running_back, 8 wide_receivers, 101.5ms
Speed: 0.9ms preprocess, 101.5ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-mp4-143_jpg.rf.4a82f8446679a5d4befd96269a4ba1b5.jpg: 640x640 13 defenses, 8 olines, 2 qbs, 3 refs, 2 running_backs, 5 wide_receivers, 101.3ms
Speed: 0.6ms preprocess, 101.3ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57724_001094_Sideline_frame124_jpg.rf.cf19beade3ef7bcc4bdd495a6dbc058c.jpg: 640x640 14 defenses, 2 olines, 3 qbs, 9 refs, 2 running_backs, 8 wide_receivers, 105.8ms
Speed: 0.7ms preprocess, 105.8ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

(1, 3, 640, 640)

Processing 81/200:

WashSt_vs_IowaSt2018_15_mp4-1_jpg.rf.6090d76e9107538f21107b85f377d11e.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/WashSt_vs_IowaSt2018_15_mp4-1_jpg.rf.6090d76e9107538f21107b85f377d11e.jpg: 640x640 16 defenses, 6 olines, 1 qb, 17 refs, 8 wide_receivers, 102.5ms
Speed: 0.7ms preprocess, 102.5ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Oregon_0_vs_Colorado_2019_mp4-53_jpg.rf.970d62b0ef00d2b212d918f3fe0c74f6.jpg: 640x640 17 defenses, 6 olines, 2 qbs, 4 refs, 2 running_backs, 3 wide_receivers, 104.8ms
Speed: 0.7ms preprocess, 104.8ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Oregon_0_vs_Colorado_2019_mp4-48_jpg.rf.9a3a4240c3a1056a6858836232b3aa82.jpg: 640x640 13 defenses, 8 olines, 3 qbs, 6 refs, 3 running_backs, 3 wide_receivers, 102.2ms
Speed: 0.7ms preprocess, 102.2ms inference, 1.3ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57604_001602_Sideline_frame193_jpg.rf.e142b1b64612f5d7179926dd2580121e.jpg: 640x640 14 defenses, 11 olines, 1 running_back, 2 wide_receivers, 102.7ms
Speed: 0.7ms preprocess, 102.7ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture128_PNG_jpg.rf.4e9a0b3bca584d1e29c9d8b7ea6d8782.jpg: 640x640 16 defenses, 4 olines, 2 qbs, 3 running_backs, 5 wide_receivers, 105.2ms
Speed: 0.7ms preprocess, 105.2ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture39_PNG_jpg.rf.48b4f1f5e1d9ed2a53999ab4f94a9362.jpg: 640x640 14 defenses, 9 olines, 2 refs, 1 running_back, 4 wide_receivers, 114.7ms
Speed: 0.7ms preprocess, 114.7ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture208_PNG_jpg.rf.a9bca8c03568f5c302

2e95d6e8ee1abb.jpg: 640x640 14 defenses, 10 olines, 4 refs, 3 wide_receivers, 103.7ms

Speed: 0.7ms preprocess, 103.7ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture127_PNG_jpg.rf.c409eb73dcd24e18245c5663896d2f4b.jpg: 640x640 12 defenses, 3 olines, 3 qbs, 1 ref, 4 running_backs, 6 wide_receivers, 97.8ms

Speed: 0.6ms preprocess, 97.8ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-156_jpg.rf.11fb8d432d62328ee3919fbe894ff55c.jpg: 640x640 20 defenses, 2 running_backs, 5 wide_receivers, 100.8ms

Speed: 0.7ms preprocess, 100.8ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture184_PNG_jpg.rf.337ff48967886e1693897f01c2814a24.jpg: 640x640 15 defenses, 4 olines, 13 qbs, 4 running_backs, 8 wide_receivers, 104.6ms

Speed: 0.6ms preprocess, 104.6ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

Processing 91/200:

58114_003195_Sideline_frame0674_jpg.rf.ce192116798a23fc42d3d2cffc8a3dd7.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58114_003195_Sideline_frame0674_jpg.rf.ce192116798a23fc42d3d2cffc8a3dd7.jpg: 640x640 18 defenses, 8 olines, 1 running_back, 2 wide_receivers, 114.5ms

Speed: 0.6ms preprocess, 114.5ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture12_PNG_jpg.rf.d5e6c94cf5d6add939c80a26da639486.jpg: 640x640 19 defenses, 4 olines, 1 qb, 1 ref, 1 running_back, 7 wide_receivers, 108.4ms

Speed: 0.8ms preprocess, 108.4ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture155_PNG_jpg.rf.b667e783c3fc315c192ea3e9afe47d5c.jpg: 640x640 13 defenses, 18 olines, 7 refs, 2 wide_receivers, 107.3ms

Speed: 0.6ms preprocess, 107.3ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Screen-Recording-2023-09-20-at-6_51_47-PM_mov-9_jpg.rf.076a163de878370af39d6ab601aaada5.jpg: 640x640 13 defenses, 6 olines, 4 qbs, 12 refs, 4 running_backs, 6 wide_receivers, 104.9ms
Speed: 0.6ms preprocess, 104.9ms inference, 2.2ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture45_PNG_jpg.rf.442f3478efca5d421cfa9b20661fbcf0.jpg: 640x640 14 defenses, 8 olines, 4 refs, 1 wide_receiver, 100.7ms
Speed: 0.9ms preprocess, 100.7ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57884_002855_Sideline_frame36_jpg.rf.edd88f1cb28f39a4f1a5157ad10e13c8.jpg: 640x640 17 defenses, 1 oline, 8 qbs, 21 refs, 1 running_back, 5 wide_receivers, 105.1ms
Speed: 0.7ms preprocess, 105.1ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture55_PNG_jpg.rf.10e65ce428fe94515a9ca2840622ac32.jpg: 640x640 13 defenses, 5 olines, 2 qbs, 3 refs, 1 running_back, 7 wide_receivers, 120.3ms
Speed: 0.7ms preprocess, 120.3ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture2_PNG_jpg.rf.4903ed7ad36f9c42fa52daeb1ad5b278.jpg: 640x640 11 defenses, 6 olines, 1 running_back, 4 wide_receivers, 102.9ms
Speed: 0.7ms preprocess, 102.9ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture129_PNG_jpg.rf.0d9bfb2ca6222e37ddb37d4258fdce0b.jpg: 640x640 16 defenses, 8 olines, 3 refs, 2 running_backs, 2 wide_receivers, 102.6ms
Speed: 0.6ms preprocess, 102.6ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-57_jpg.rf.817a1c3d67700a42dea90f9414363a95.jpg: 640x640 15 defenses, 3 olines, 6 wide_receivers, 100.7ms
Speed: 0.7ms preprocess, 100.7ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

(1, 3, 640, 640)

Processing 101/200: yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-137_jpg.rf.b8e9025d8bb4d12c613e3c6e7cc67c24.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-137_jpg.rf.b8e9025d8bb4d12c613e3c6e7cc67c24.jpg: 640x640 18 defenses, 7 olines, 15 refs, 4 wide_receivers, 99.7ms
Speed: 0.7ms preprocess, 99.7ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57506_000803_Sideline_frame1054_jpg.rf.01e9c6addaadb9da7482049e98eda867.jpg: 640x640 16 defenses, 9 olines, 2 refs, 5 wide_receivers, 101.6ms
Speed: 0.7ms preprocess, 101.6ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Sideline3_mp4-0_jpg.rf.ee74bc1a00cbc584209224e27537764d.jpg: 640x640 17 defenses, 2 olines, 1 qb, 7 refs, 4 wide_receivers, 101.0ms
Speed: 0.6ms preprocess, 101.0ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture177_PNG_jpg.rf.67b8cb2faa6e838c9ccb2219dcd84b39.jpg: 640x640 17 defenses, 7 olines, 2 refs, 2 wide_receivers, 104.5ms
Speed: 0.6ms preprocess, 104.5ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-57_jpg.rf.0760d06a49dcde9d5042c414bb674185.jpg: 640x640 14 defenses, 2 olines, 8 qbs, 11 refs, 3 running_backs, 5 wide_receivers, 101.1ms
Speed: 0.6ms preprocess, 101.1ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Sideline5_mp4-0_jpg.rf.027fc397e405a9ab30dea8402b4729bd.jpg: 640x640 13 defenses, 2 olines, 8 qbs, 1 ref, 2 running_backs, 6 wide_receivers, 104.9ms
Speed: 0.6ms preprocess, 104.9ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/

notebooks/../../data/original/train/images/Sideline9_mp4-0_jpg.rf.7c9692426155ef6ea1829137a1a0c78d.jpg: 640x640 12 defenses, 6 olines, 3 qbs, 2 refs, 4 running_backs, 4 wide_receivers, 141.1ms
Speed: 0.6ms preprocess, 141.1ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/58157_001501_Sideline_frame140_jpg.rf.422bc313605d6a627966b2d624998eea.jpg: 640x640 19 defenses, 5 olines, 4 refs, 1 running_back, 5 wide_receivers, 112.2ms
Speed: 0.6ms preprocess, 112.2ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57890_000485_Sideline_frame93_jpg.rf.83c98f92df3384b7a0c3b4d64e6ca46b.jpg: 640x640 15 defenses, 5 olines, 2 qbs, 4 refs, 5 wide_receivers, 107.1ms
Speed: 0.7ms preprocess, 107.1ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture66_PNG_jpg.rf.4f7fef2487bae30837a3e669d642aa32.jpg: 640x640 18 defenses, 2 olines, 1 ref, 3 wide_receivers, 102.5ms
Speed: 0.6ms preprocess, 102.5ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

Processing 111/200:

WashSt_vs_IowaSt2018_mp4-78_jpg.rf.63af9e20a5dfe406c7fecfef8c3d0df6.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/WashSt_vs_IowaSt2018_mp4-78_jpg.rf.63af9e20a5dfe406c7fecfef8c3d0df6.jpg: 640x640 16 defenses, 7 olines, 5 qbs, 17 refs, 5 wide_receivers, 111.4ms
Speed: 0.6ms preprocess, 111.4ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57907_003078_Sideline_frame31_jpg.rf.8a33e2faa89fe49e217704ec74a05b54.jpg: 640x640 13 defenses, 13 olines, 1 qb, 1 ref, 1 running_back, 8 wide_receivers, 103.3ms
Speed: 0.6ms preprocess, 103.3ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57570_000107_Sideline_frame108_jpg.rf.cc0ebd76344a5c76402ea28fbfc61f3.jpg: 640x640 17 defenses, 6 olines, 13 refs, 4 wide_receivers, 110.2ms
Speed: 0.6ms preprocess, 110.2ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture3_PNG_jpg.rf.d7db23424cb33403e8cd3ea2d415ad01.jpg: 640x640 20 defenses, 6 olines, 5 qbs, 8 refs, 5 running_backs, 3 wide_receivers, 105.3ms
Speed: 0.7ms preprocess, 105.3ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture200_PNG_jpg.rf.2d67477e83019c553eb7372c0d7bfdaa.jpg: 640x640 14 defenses, 6 olines, 1 qb, 1 ref, 1 running_back, 1 tight_end, 4 wide_receivers, 101.9ms
Speed: 0.7ms preprocess, 101.9ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture39_PNG_jpg.rf.99e4d8dab4b7ac00818bfc9682491361.jpg: 640x640 15 defenses, 6 olines, 4 refs, 1 running_back, 2 wide_receivers, 101.2ms
Speed: 0.6ms preprocess, 101.2ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58117_002168_Sideline_frame22_jpg.rf.98973465bd742375601e7f39e21306de.jpg: 640x640 17 defenses, 4 olines, 3 refs, 1 running_back, 3 wide_receivers, 102.1ms
Speed: 0.8ms preprocess, 102.1ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57679_001946_Sideline_frame46_jpg.rf.8492f9850d88784591235d95b58aab94.jpg: 640x640 18 defenses, 6 olines, 1 qb, 4 refs, 1 running_back, 4 wide_receivers, 103.5ms
Speed: 0.7ms preprocess, 103.5ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture_PNG_jpg.rf.df55555f222ce86a2204ccb8391796f1.jpg: 640x640 12 defenses, 7 olines, 2 qbs, 2 running_backs, 4 wide_receivers, 102.2ms
Speed: 0.7ms preprocess, 102.2ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57538_000160_Sideline_frame14_jpg.rf.894b7ee5a4a2d8306718387012a14b0b.jpg: 640x640 17 defenses, 8 olines, 1 qb, 3 refs, 1 running_back, 4 wide_receivers, 101.2ms

Speed: 0.7ms preprocess, 101.2ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

Processing 121/200:

58061_001918_Sideline_frame0836_jpg.rf.1967d3bdd2f60456d3db110d1c24e6e9.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58061_001918_Sideline_frame0836_jpg.rf.1967d3bdd2f60456d3db110d1c24e6e9.jpg: 640x640 17 defenses, 1 oline, 1 ref, 6 wide_receivers, 101.5ms

Speed: 0.6ms preprocess, 101.5ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture_PNG_jpg.rf.62d6a0d2e45727c300e307be878e2e0e.jpg: 640x640 14 defenses, 7 olines, 1 ref, 4 wide_receivers, 105.1ms
Speed: 0.7ms preprocess, 105.1ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58005_003391_Sideline_frame134_jpg.rf.1517939b1dadcafe28e4a8ec541bbd0e.jpg: 640x640 19 defenses, 2 qbs, 21 refs, 4 wide_receivers, 102.6ms
Speed: 0.6ms preprocess, 102.6ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture20_PNG_jpg.rf.c507c1115c6808a019d462da95869732.jpg: 640x640 17 defenses, 8 olines, 4 refs, 104.2ms
Speed: 0.7ms preprocess, 104.2ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58036_003522_Sideline_frame99_jpg.rf.05b5fa79ac80ed9fe6848f0c67e6a82e.jpg: 640x640 16 defenses, 5 olines, 1 qb, 5 refs, 1 running_back, 5 wide_receivers, 101.9ms
Speed: 0.6ms preprocess, 101.9ms inference, 0.4ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture96_PNG_jpg.rf.3d261d1391f950ce594ff9338bf283d0.jpg: 640x640 16 defenses, 5 olines, 1 qb, 3 refs, 2 running_backs, 5 wide_receivers, 101.4ms
Speed: 0.7ms preprocess, 101.4ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Oregon_0_vs_Colorado_2019_mp4-89_jpg.rf.bb4e4e39f3035dc65f98e51488f4fed2.jpg: 640x640 15 defenses, 2 qbs, 13

refs, 1 running_back, 6 wide_receivers, 100.8ms
Speed: 0.6ms preprocess, 100.8ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/WashSt_vs_IowaSt2018_mp4-
78_jpg.rf.f08e895733368b65d6c90d451528f983.jpg: 640x640 13 defenses, 6 olines, 2
qbs, 12 refs, 1 running_back, 7 wide_receivers, 103.3ms
Speed: 0.6ms preprocess, 103.3ms inference, 0.9ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/Capture19_PNG_jpg.rf.f585fbbc2e2c9ec38d0
508c56505d6c7.jpg: 640x640 12 defenses, 6 olines, 4 refs, 2 running_backs, 7
wide_receivers, 104.3ms
Speed: 0.6ms preprocess, 104.3ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/57937_003316_Sideline_frame0663_jpg.rf.3
0b2c99f06ae3d68dd68931389bb7c14.jpg: 640x640 17 defenses, 6 olines, 1 qb, 4
refs, 1 running_back, 5 wide_receivers, 104.5ms
Speed: 0.7ms preprocess, 104.5ms inference, 0.6ms postprocess per image at shape
(1, 3, 640, 640)

Processing 131/200:
Capture219_PNG_jpg.rf.a6aaeb8b2adf891a0defe1fe33a506f0.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/Capture219_PNG_jpg.rf.a6aaeb8b2adf891a0d
efe1fe33a506f0.jpg: 640x640 13 defenses, 6 olines, 2 refs, 3 running_backs, 5
wide_receivers, 109.3ms
Speed: 0.7ms preprocess, 109.3ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/57681_001689_Sideline_frame94_jpg.rf.d1d
c7ab9552a5d03ab3a62ed58c646fe.jpg: 640x640 17 defenses, 3 olines, 2 qbs, 2
running_backs, 9 wide_receivers, 113.5ms
Speed: 0.7ms preprocess, 113.5ms inference, 0.6ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/Capture12_PNG_jpg.rf.2e5a4cae62bcfa3aad5
4db8e3d244a7b.jpg: 640x640 15 defenses, 8 olines, 1 qb, 1 running_back, 8
wide_receivers, 105.2ms
Speed: 0.7ms preprocess, 105.2ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture9_PNG_jpg.rf.d8e25b9063ae84430d47d18fe26faf19.jpg: 640x640 16 defenses, 6 qbs, 16 refs, 6 running_backs, 1 wide_receiver, 104.4ms
Speed: 0.6ms preprocess, 104.4ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/58136_002740_Sideline_frame215_jpg.rf.27afcd8a65cbab5b1ca15a59e1eb23be.jpg: 640x640 14 defenses, 4 olines, 1 qb, 1 running_back, 4 wide_receivers, 102.1ms
Speed: 0.7ms preprocess, 102.1ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57693_002948_Sideline_frame159_jpg.rf.4f7234f3f1333208484cb265f3f8d8ea.jpg: 640x640 15 defenses, 4 olines, 3 qbs, 7 refs, 4 running_backs, 3 wide_receivers, 101.5ms
Speed: 0.6ms preprocess, 101.5ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/cap-2-_png_jpg.rf.27d67a088d50e8ab96ea6a0d867cb8ef.jpg: 640x640 16 defenses, 6 olines, 1 ref, 3 wide_receivers, 103.1ms
Speed: 0.6ms preprocess, 103.1ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture213_PNG_jpg.rf.519895176a01a32da52fd16ee7991dc7.jpg: 640x640 14 defenses, 5 olines, 4 wide_receivers, 105.2ms
Speed: 0.7ms preprocess, 105.2ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-72_jpg.rf.d3f3acd07db8b305b41f8a52c7aed1ab.jpg: 640x640 18 defenses, 2 olines, 1 qb, 3 refs, 2 running_backs, 5 wide_receivers, 134.6ms
Speed: 0.6ms preprocess, 134.6ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57558_002886_Sideline_frame204_jpg.rf.386300f508e1beea9a6034e14cb6fca8.jpg: 640x640 15 defenses, 4 olines, 1 qb, 1 ref, 2 running_backs, 5 wide_receivers, 103.8ms
Speed: 0.7ms preprocess, 103.8ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

Processing 141/200:
WashSt_vs_IowaSt2018_mp4-58_jpg.rf.6d8d40654aebab2f6b9d4424d4740697.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/WashSt_vs_IowaSt2018_mp4-58_jpg.rf.6d8d40654aebab2f6b9d4424d4740697.jpg: 640x640 15 defenses, 5 olines, 1 qb, 6 refs, 1 running_back, 7 wide_receivers, 102.0ms
Speed: 0.6ms preprocess, 102.0ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57686_000570_Sideline_frame158_jpg.rf.d100142b6917b4faa1f473b56d58dfc5.jpg: 640x640 15 defenses, 7 olines, 4 wide_receivers, 106.5ms
Speed: 0.7ms preprocess, 106.5ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture19_PNG_jpg.rf.58abae12e3c5d906f35c346285d47e1a.jpg: 640x640 12 defenses, 3 olines, 5 qbs, 2 running_backs, 4 wide_receivers, 105.6ms
Speed: 0.6ms preprocess, 105.6ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57870_000298_Sideline_frame185_jpg.rf.1e0c2a4685c5a43f2c3afa8989a2485d.jpg: 640x640 17 defenses, 8 olines, 1 qb, 2 running_backs, 5 wide_receivers, 101.2ms
Speed: 0.6ms preprocess, 101.2ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture125_PNG_jpg.rf.91cdafc2b15309bb1e1c6ad570495395.jpg: 640x640 16 defenses, 5 olines, 3 refs, 3 running_backs, 5 wide_receivers, 104.4ms
Speed: 0.7ms preprocess, 104.4ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture16_PNG_jpg.rf.ea6047063dc2c387f30caa721eb29a89.jpg: 640x640 12 defenses, 6 olines, 4 qbs, 4 running_backs, 2 wide_receivers, 100.3ms
Speed: 0.6ms preprocess, 100.3ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Oregon_0_vs_Colorado_2019_mp4-85_jpg.rf.31688aff199a5504e668fb4a9ef1b56a.jpg: 640x640 16 defenses, 6 olines, 6

refs, 1 running_back, 4 wide_receivers, 103.7ms
Speed: 0.6ms preprocess, 103.7ms inference, 1.0ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/57919_002198_Sideline_frame68_jpg.rf.ea6
7077afbc9b7d02dd5067afbf14cb1.jpg: 640x640 15 defenses, 7 olines, 1 qb, 9 refs,
1 running_back, 10 wide_receivers, 99.9ms
Speed: 0.6ms preprocess, 99.9ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/Capture126_PNG_jpg.rf.a0c88225e04eae5134
45b227783949a6.jpg: 640x640 13 defenses, 5 olines, 1 ref, 3 wide_receivers,
179.1ms
Speed: 0.7ms preprocess, 179.1ms inference, 1.1ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/58162_000613_Sideline_frame1135_jpg.rf.d
2dd0cf361d7309ee72fbe980a6e1a0b.jpg: 640x640 14 defenses, 5 olines, 3 qbs, 8
refs, 4 running_backs, 4 wide_receivers, 137.0ms
Speed: 2.7ms preprocess, 137.0ms inference, 0.6ms postprocess per image at shape
(1, 3, 640, 640)

Processing 151/200:
57614_000361_Sideline_frame60_jpg.rf.5c8adab2f4957416636fc22054c06f7d.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/57614_000361_Sideline_frame60_jpg.rf.5c8
adab2f4957416636fc22054c06f7d.jpg: 640x640 16 defenses, 6 olines, 1
wide_receiver, 112.7ms
Speed: 0.6ms preprocess, 112.7ms inference, 0.9ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/Capture8_PNG_jpg.rf.79975bd751048518c4f1
32573041a8ca.jpg: 640x640 16 defenses, 5 olines, 1 qb, 2 running_backs, 7
wide_receivers, 106.7ms
Speed: 0.7ms preprocess, 106.7ms inference, 0.9ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/../../data/original/train/images/57936_002253_Sideline_frame0559_jpg.rf.5
3ce1c540fb75e9c28378028478f74b6.jpg: 640x640 20 defenses, 3 olines, 1 ref, 2
wide_receivers, 102.4ms
Speed: 0.6ms preprocess, 102.4ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/WashSt_vs_IowaSt2018_mp4-92_jpg.rf.1ddcaaf5f0703d16f191ddd3392fb228.jpg: 640x640 15 defenses, 8 qbs, 9 refs, 3 running_backs, 7 wide_receivers, 101.9ms
Speed: 0.6ms preprocess, 101.9ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57946_001177_Sideline_frame0844_jpg.rf.0321fcc9a8dfb3e79f8da523304eec6c.jpg: 640x640 15 defenses, 4 olines, 4 wide_receivers, 112.1ms
Speed: 0.6ms preprocess, 112.1ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/yt5s_com-Detroit-Lions-vs-Dallas-Cowboys-2016-All-22-Film-360p-_mp4-120_jpg.rf.4c1ccf5890457a4cfbd68e6c7deace3f.jpg: 640x640 16 defenses, 6 olines, 2 qbs, 2 refs, 2 running_backs, 5 wide_receivers, 102.3ms
Speed: 0.6ms preprocess, 102.3ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture17_PNG_jpg.rf.b4417246706a8c99b9789b9422b2fe34.jpg: 640x640 14 defenses, 3 olines, 6 qbs, 4 refs, 2 running_backs, 5 wide_receivers, 113.2ms
Speed: 0.6ms preprocess, 113.2ms inference, 1.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture2_PNG_jpg.rf.77efaa5501d35cc418e2b8376381f619.jpg: 640x640 16 defenses, 4 olines, 1 running_back, 3 wide_receivers, 164.2ms
Speed: 0.9ms preprocess, 164.2ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57673_001856_Sideline_frame4_jpg.rf.868ec8b1515239c7f92e924f1abae822.jpg: 640x640 18 defenses, 5 olines, 1 ref, 1 running_back, 5 wide_receivers, 127.5ms
Speed: 1.8ms preprocess, 127.5ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/videoautoplayback_mp4-139_jpg.rf.3805f13eb609d70f18b3225290002a2b.jpg: 640x640 15 defenses, 6 olines, 4 refs, 2 running_backs, 2 wide_receivers, 240.7ms
Speed: 0.7ms preprocess, 240.7ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

(1, 3, 640, 640)

Processing 161/200:

57984_001874_Sideline_frame18_jpg.rf.ce7dc5e4037f13d4c054d8d28ffea3fb.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57984_001874_Sideline_frame18_jpg.rf.ce7dc5e4037f13d4c054d8d28ffea3fb.jpg: 640x640 17 defenses, 2 olines, 4 qbs, 4 refs, 1 running_back, 6 wide_receivers, 116.4ms

Speed: 0.7ms preprocess, 116.4ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57630_004303_Sideline_frame58_jpg.rf.b352f4232f72307744a9500f03820c9b.jpg: 640x640 16 defenses, 5 olines, 8 refs, 1 running_back, 4 wide_receivers, 115.7ms

Speed: 0.6ms preprocess, 115.7ms inference, 1.4ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Bengals_mp4-2_jpg.rf.92a52261f9b73520575a3716c2dec547.jpg: 640x640 12 defenses, 5 olines, 2 qbs, 1 ref, 3 running_backs, 5 wide_receivers, 140.4ms

Speed: 0.9ms preprocess, 140.4ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture166_PNG_jpg.rf.897ffdd3ce21f40ffb4b4a00f3257c65.jpg: 640x640 15 defenses, 8 olines, 1 qb, 2 refs, 3 running_backs, 4 wide_receivers, 169.2ms

Speed: 0.8ms preprocess, 169.2ms inference, 1.4ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture31_PNG_jpg.rf.acce311543ab63b2c2fe35b5738addcd.jpg: 640x640 18 defenses, 2 olines, 1 qb, 5 refs, 3 running_backs, 4 wide_receivers, 121.3ms

Speed: 0.9ms preprocess, 121.3ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/videoautoplayback_mp4-149_jpg.rf.09d0ca1115c121dbbaed643f7338fb94.jpg: 640x640 11 defenses, 6 olines, 3 qbs, 2 refs, 2 running_backs, 8 wide_receivers, 103.5ms

Speed: 0.6ms preprocess, 103.5ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57594_004453_Sideline_frame102_jpg.rf.b0

d0a30f67d5f062ac4f0240681ada96.jpg: 640x640 12 defenses, 7 olines, 5 qbs, 2 refs, 4 running_backs, 5 wide_receivers, 130.2ms
Speed: 0.7ms preprocess, 130.2ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57885_002966_Sideline_frame23_jpg.rf.bd9d7a84ff3ad5c93a728dc57cf95fd1.jpg: 640x640 16 defenses, 1 oline, 4 qbs, 12 refs, 1 running_back, 7 wide_receivers, 122.0ms
Speed: 0.6ms preprocess, 122.0ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58013_003447_Sideline_frame0640_jpg.rf.03d61687fba960a8e64a72da7e2de1d9.jpg: 640x640 12 defenses, 4 olines, 1 qb, 4 refs, 1 running_back, 7 wide_receivers, 150.9ms
Speed: 0.7ms preprocess, 150.9ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture122_PNG_jpg.rf.ea68785e585b0762c297213c8be5b902.jpg: 640x640 13 defenses, 5 olines, 3 qbs, 1 ref, 1 running_back, 3 wide_receivers, 103.1ms
Speed: 0.6ms preprocess, 103.1ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

Processing 171/200:

58052_003602_Sideline_frame116_jpg.rf.7e0f2591a1198be61a5c7ed2c97744ec.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58052_003602_Sideline_frame116_jpg.rf.7e0f2591a1198be61a5c7ed2c97744ec.jpg: 640x640 17 defenses, 6 olines, 1 qb, 8 refs, 3 running_backs, 5 wide_receivers, 104.5ms
Speed: 0.6ms preprocess, 104.5ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/2_PNG_jpg.rf.996901b08a77917194754d0a9314ec70.jpg: 640x640 13 defenses, 4 olines, 3 qbs, 1 ref, 2 running_backs, 5 wide_receivers, 104.2ms
Speed: 0.7ms preprocess, 104.2ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57515_001257_Sideline_frame22_jpg.rf.24d28432b60e8aa42e60cff535637048.jpg: 640x640 13 defenses, 6 olines, 2 refs, 4 wide_receivers, 109.8ms
Speed: 0.7ms preprocess, 109.8ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture156_PNG_jpg.rf.86643f75f047f148e7b6ad94304cd236.jpg: 640x640 13 defenses, 7 olines, 2 qbs, 1 ref, 2 running_backs, 5 wide_receivers, 112.2ms
Speed: 0.7ms preprocess, 112.2ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57893_002454_Sideline_frame106_jpg.rf.15a118a78cbf8ba1b2b161667337ce81.jpg: 640x640 12 defenses, 4 olines, 2 qbs, 4 running_backs, 4 wide_receivers, 114.6ms
Speed: 0.7ms preprocess, 114.6ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture10_PNG_jpg.rf.8c8cedfb415638480faf750235f03617.jpg: 640x640 18 defenses, 6 olines, 2 refs, 1 running_back, 5 wide_receivers, 107.0ms
Speed: 0.8ms preprocess, 107.0ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57972_001431_Sideline_frame194_jpg.rf.ec95fa66273d6c604d28f89d2be3b727.jpg: 640x640 18 defenses, 4 olines, 6 wide_receivers, 108.5ms
Speed: 0.8ms preprocess, 108.5ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture231_PNG_jpg.rf.3416f4a4000088d8e6f970b8abac97d6.jpg: 640x640 16 defenses, 7 olines, 1 qb, 1 ref, 1 running_back, 5 wide_receivers, 104.1ms
Speed: 0.6ms preprocess, 104.1ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58095_002347_Sideline_frame107_jpg.rf.8c6f3ab5e0f4c205aae5a91285f49167.jpg: 640x640 18 defenses, 6 olines, 1 qb, 10 refs, 3 running_backs, 3 wide_receivers, 109.4ms
Speed: 0.7ms preprocess, 109.4ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-mp4-5_jpg.rf.05b99f4d7fa9a72b43eef3ca636c883e.jpg: 640x640 14 defenses, 6 olines, 1 qb, 7 refs, 3 running_backs, 4 wide_receivers, 108.5ms

Speed: 0.8ms preprocess, 108.5ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

Processing 181/200: Captur3_PNG_jpg.rf.d650d26a11dfbe24672eea326db4bfef.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Captur3_PNG_jpg.rf.d650d26a11dfbe24672eea326db4bfef.jpg: 640x640 21 defenses, 6 olines, 2 qbs, 3 refs, 1 running_back, 5 wide_receivers, 124.1ms

Speed: 0.6ms preprocess, 124.1ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture16_PNG_jpg.rf.8d6bdaca4d9cfd8d736f101c2db411a0.jpg: 640x640 16 defenses, 8 olines, 2 refs, 2 running_backs, 5 wide_receivers, 103.3ms

Speed: 0.6ms preprocess, 103.3ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57728_001817_Sideline_frame0763_jpg.rf.53b00573614263377921dd2e253f304c.jpg: 640x640 13 defenses, 6 olines, 1 qb, 2 refs, 2 running_backs, 3 wide_receivers, 104.8ms

Speed: 0.7ms preprocess, 104.8ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/58113_000424_Sideline_frame0968_jpg.rf.3d25d8aadd80bffe051421ef3060da10.jpg: 640x640 19 defenses, 3 olines, 3 refs, 2 wide_receivers, 115.8ms

Speed: 0.7ms preprocess, 115.8ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture14_PNG_jpg.rf.971a9ef0b0db33be748c32b2354ac882.jpg: 640x640 15 defenses, 3 olines, 6 qbs, 2 refs, 2 running_backs, 5 wide_receivers, 101.4ms

Speed: 0.6ms preprocess, 101.4ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture221_PNG_jpg.rf.40bc4fcab8ecd8fb9dc94bdcc3ff7a07.jpg: 640x640 16 defenses, 5 olines, 1 qb, 1 ref, 1 running_back, 8 wide_receivers, 105.1ms

Speed: 0.7ms preprocess, 105.1ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture241_PNG_jpg.rf.43dffb4aa31589736d

ab9c81ae3947d3.jpg: 640x640 17 defenses, 4 olines, 1 qb, 1 running_back, 5 wide_receivers, 102.7ms
Speed: 0.7ms preprocess, 102.7ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture201_PNG_jpg.rf.0e61872fc9115644c9474e7c70a698d3.jpg: 640x640 18 defenses, 7 olines, 4 qbs, 1 ref, 3 wide_receivers, 111.7ms
Speed: 0.6ms preprocess, 111.7ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58060_002645_Sideline_frame84_jpg.rf.f56dae6ed766e6c5caaea57aae3acb47.jpg: 640x640 19 defenses, 4 olines, 7 wide_receivers, 104.3ms
Speed: 0.7ms preprocess, 104.3ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/58163_000542_Sideline_frame0662_jpg.rf.11de26c2f1def8ed5a34445b5a9a74c4.jpg: 640x640 15 defenses, 5 olines, 1 ref, 2 running_backs, 2 wide_receivers, 102.1ms
Speed: 0.6ms preprocess, 102.1ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

Processing 191/200: yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-130_jpg.rf.50691a0fa33a203b0e5871923e14c504.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-130_jpg.rf.50691a0fa33a203b0e5871923e14c504.jpg: 640x640 11 defenses, 7 olines, 3 qbs, 1 ref, 2 running_backs, 7 wide_receivers, 112.5ms
Speed: 0.7ms preprocess, 112.5ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57932_002365_Sideline_frame0836_jpg.rf.11884833bc33411d5f902bcf38d8b0e7.jpg: 640x640 16 defenses, 6 olines, 1 ref, 2 running_backs, 3 wide_receivers, 100.9ms
Speed: 0.7ms preprocess, 100.9ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/Capture18_PNG_jpg.rf.ec2b3d6531413bba816fc21e01c8c4e3.jpg: 640x640 11 defenses, 5 olines, 1 ref, 1 running_back, 3 wide_receivers, 101.5ms
Speed: 0.6ms preprocess, 101.5ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture184_PNG_jpg.rf.6062a12370320486b8e220c936edb80e.jpg: 640x640 15 defenses, 5 olines, 3 qbs, 2 running_backs, 5 wide_receivers, 104.1ms
Speed: 0.6ms preprocess, 104.1ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/Capture9_PNG_jpg.rf.6252c04b463074f3404ba446bc82bb7d.jpg: 640x640 15 defenses, 6 olines, 3 qbs, 15 refs, 3 running_backs, 2 wide_receivers, 102.4ms
Speed: 0.6ms preprocess, 102.4ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/58152_002751_Sideline_frame171_jpg.rf.6c05f18db2971222d6d029bdd9df2031.jpg: 640x640 12 defenses, 1 oline, 19 qbs, 11 running_backs, 4 wide_receivers, 102.9ms
Speed: 0.6ms preprocess, 102.9ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57770_001936_Sideline_frame40_jpg.rf.ffa4c405f5fee7b0c8077e92b589a190.jpg: 640x640 14 defenses, 1 oline, 3 qbs, 1 ref, 2 running_backs, 4 wide_receivers, 154.2ms
Speed: 0.7ms preprocess, 154.2ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/58114_003195_Sideline_frame0674_jpg.rf.d300ec334cf0a5c8ced75a8ef9f0e938.jpg: 640x640 18 defenses, 4 olines, 3 running_backs, 3 wide_receivers, 105.8ms
Speed: 0.7ms preprocess, 105.8ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/yt5s_com-Super-Bowl-LV-All-22-Coaches-Film-480p-mp4-29_jpg.rf.5b15705f660ba4251de10767221aa170.jpg: 640x640 13 defenses, 8 olines, 4 qbs, 3 refs, 2 running_backs, 2 wide_receivers, 112.5ms
Speed: 0.7ms preprocess, 112.5ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/train/images/57862_000058_Sideline_frame35_jpg.rf.70da79b01d8ef8f6e0c0bc1e8e275e36.jpg: 640x640 11 defenses, 7 olines, 1 qb, 1 ref, 1

running_back, 3 wide_receivers, 99.3ms
Speed: 0.6ms preprocess, 99.3ms inference, 0.5ms postprocess per image at shape
(1, 3, 640, 640)
Successfully processed: 200 images
Failed extractions: 0 images

ALTERNATIVE VALID FOLDER

Processing original_valid (valid) folder...

Found 50 images

Processing 1/50: Capture23_PNG_jpg.rf.ccf91081aa07b3f2464b1af17ce6097c.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/./data/original/valid/images/Capture23_PNG_jpg.rf.ccf91081aa07b3f2464
b1af17ce6097c.jpg: 640x640 13 defenses, 1 oline, 13 qbs, 7 refs, 3
running_backs, 6 wide_receivers, 103.5ms
Speed: 0.6ms preprocess, 103.5ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/./data/original/valid/images/Capture54_PNG_jpg.rf.a2fd8b0527fd19a74cf
b21997c1a4297.jpg: 640x640 12 defenses, 6 olines, 4 qbs, 1 ref, 5 running_backs,
4 wide_receivers, 102.5ms
Speed: 0.7ms preprocess, 102.5ms inference, 0.6ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/
notebooks/./data/original/valid/images/57643_002968_Sideline_frame1188_jpg.rf.c
285923e45303656461ac8963d3ae2bf.jpg: 640x640 13 defenses, 4 qbs, 12 refs, 2
running_backs, 10 wide_receivers, 103.1ms
Speed: 0.6ms preprocess, 103.1ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-
ai/notebooks/./data/original/valid/images/yt5s_com-Detroit-Lions-vs-Dallas-Cowb
oys-2016-All-22-Film-360p-_mp4-107_jpg.rf.78a08bd580cd5d40fff9dc2bf3ad6949.jpg:
640x640 18 defenses, 3 olines, 5 refs, 7 wide_receivers, 103.1ms
Speed: 0.6ms preprocess, 103.1ms inference, 0.9ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-
ai/notebooks/./data/original/valid/images/yt5s_com-Vikings-2017-3rd-Down-
Defense-All-22-Cut-up-480p-_mp4-177_jpg.rf.dc3705fbd5b40f1dade6d2c049cbf38d.jpg:
640x640 12 defenses, 8 qbs, 1 ref, 3 running_backs, 4 wide_receivers, 102.4ms
Speed: 0.7ms preprocess, 102.4ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/

notebooks/../../data/original/valid/images/Capture22_PNG_jpg.rf.a3e8e1d93dab8edc50476e71a5045fc4.jpg: 640x640 15 defenses, 4 olines, 2 qbs, 19 refs, 2 running_backs, 6 wide_receivers, 104.4ms
Speed: 0.7ms preprocess, 104.4ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/58127_001780_Sideline_frame85_jpg.rf.fc321598f243ba8dc431f04849bca7c8.jpg: 640x640 21 defenses, 5 olines, 8 qbs, 3 refs, 4 running_backs, 3 wide_receivers, 101.6ms
Speed: 0.6ms preprocess, 101.6ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/Capture42_PNG_jpg.rf.c0fb5d0f8b11c3cf5fb931a4da9de594.jpg: 640x640 21 defenses, 6 olines, 1 qb, 7 refs, 1 running_back, 8 wide_receivers, 103.5ms
Speed: 0.6ms preprocess, 103.5ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/Capture1_PNG_jpg.rf.f4001272d6214bd203f73f4fed4d9359.jpg: 640x640 14 defenses, 1 oline, 8 qbs, 7 refs, 7 running_backs, 7 wide_receivers, 103.8ms
Speed: 0.7ms preprocess, 103.8ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/videoplayback_mp4-158_jpg.rf.e524328b3c6c05d3c3b0079bb0efc6bd.jpg: 640x640 13 defenses, 9 qbs, 12 refs, 1 running_back, 4 wide_receivers, 105.5ms
Speed: 0.6ms preprocess, 105.5ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

Processing 11/50: Capture165_PNG_jpg.rf.2998d73caf82cd0794144b1a4140c9e6.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/Capture165_PNG_jpg.rf.2998d73caf82cd0794144b1a4140c9e6.jpg: 640x640 12 defenses, 3 olines, 2 qbs, 7 refs, 1 running_back, 2 wide_receivers, 100.5ms
Speed: 0.7ms preprocess, 100.5ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/Capture237_PNG_jpg.rf.7cef3d57c172ff6d52b5c75d7060e6ba.jpg: 640x640 13 defenses, 8 olines, 5 qbs, 4 running_backs, 5 wide_receivers, 103.3ms
Speed: 0.7ms preprocess, 103.3ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/57897_001214_Sideline_frame157_jpg.rf.01abe0a4cb23be8aeb0b2a5eeecccae0.jpg: 640x640 12 defenses, 5 olines, 2 qbs, 1 ref, 3 running_backs, 2 wide_receivers, 111.4ms
Speed: 0.6ms preprocess, 111.4ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture153_PNG_jpg.rf.8b893825c65cc8455500eea10e0410dd.jpg: 640x640 10 defenses, 6 olines, 11 qbs, 8 refs, 1 running_back, 8 wide_receivers, 103.8ms
Speed: 0.7ms preprocess, 103.8ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture229_PNG_jpg.rf.3695979dd95dbbbb0cbf8a0e500be839.jpg: 640x640 10 defenses, 6 olines, 2 qbs, 1 ref, 2 running_backs, 4 wide_receivers, 106.7ms
Speed: 0.6ms preprocess, 106.7ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture216_PNG_jpg.rf.166d3f4e8e1216fa85494ab6ae41c24d.jpg: 640x640 12 defenses, 8 olines, 11 qbs, 1 ref, 8 running_backs, 5 wide_receivers, 105.8ms
Speed: 0.7ms preprocess, 105.8ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/57907_001496_Sideline_frame58_jpg.rf.91bd32c6175f1dd495363bbae5dc0885.jpg: 640x640 12 defenses, 5 olines, 3 qbs, 2 refs, 2 running_backs, 4 wide_receivers, 104.8ms
Speed: 0.7ms preprocess, 104.8ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture52_PNG_jpg.rf.7b73c6a7418fd1878d13d31369665837.jpg: 640x640 12 defenses, 6 olines, 9 qbs, 2 refs, 3 running_backs, 3 wide_receivers, 101.6ms
Speed: 0.6ms preprocess, 101.6ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/yt5s_com-Super-Bowl-LV-All-22-Coaches-Film-480p-mp4-54_jpg.rf.9a7e475e733d6b4e1b95854aa03bd3fb.jpg: 640x640 11 defenses, 5 olines, 9 qbs, 2 refs, 4 running_backs, 6 wide_receivers, 102.9ms

Speed: 0.7ms preprocess, 102.9ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture70_PNG_jpg.rf.0cfe570379ee40fe73e1f4a573117f07.jpg: 640x640 13 defenses, 4 olines, 3 qbs, 4 refs, 3 running_backs, 3 wide_receivers, 106.2ms
Speed: 0.7ms preprocess, 106.2ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

Processing 21/50: Capture142_PNG_jpg.rf.143e86f043913f50cae7b1c7046b3a26.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture142_PNG_jpg.rf.143e86f043913f50cae7b1c7046b3a26.jpg: 640x640 13 defenses, 14 olines, 1 qb, 10 refs, 1 running_back, 4 wide_receivers, 102.3ms
Speed: 0.7ms preprocess, 102.3ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-mp4-166_jpg.rf.66b9432779cea9f4e7691a91bdd65967.jpg: 640x640 11 defenses, 7 olines, 4 qbs, 1 ref, 5 running_backs, 4 wide_receivers, 104.1ms
Speed: 0.6ms preprocess, 104.1ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/58066_002211_Sideline_frame0746_jpg.rf.1294f5b3870b42919588deaa768c1ac0.jpg: 640x640 13 defenses, 4 olines, 5 qbs, 2 running_backs, 6 wide_receivers, 103.5ms
Speed: 0.6ms preprocess, 103.5ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture88_PNG_jpg.rf.df2a7c50333df7b3ba4af69906223e08.jpg: 640x640 18 defenses, 5 olines, 4 refs, 1 running_back, 6 wide_receivers, 142.9ms
Speed: 0.6ms preprocess, 142.9ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/57969_000592_Sideline_frame103_jpg.rf.df0e16a15d5bd828bfd9ef5f3a1483d0.jpg: 640x640 12 defenses, 6 qbs, 6 refs, 3 running_backs, 5 wide_receivers, 112.0ms
Speed: 0.7ms preprocess, 112.0ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/

notebooks/../../data/original/valid/images/Capture5_PNG_jpg.rf.e70754812ae8d9a9aa423c2854c292bc.jpg: 640x640 14 defenses, 3 olines, 7 qbs, 5 running_backs, 6 wide_receivers, 107.4ms

Speed: 0.8ms preprocess, 107.4ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-mp4-4_jpg.rf.1f91398a8abe7cf954a561b40651970b.jpg: 640x640 19 defenses, 1 qb, 10 refs, 5 wide_receivers, 110.4ms

Speed: 0.7ms preprocess, 110.4ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/yt5s_com-Super-Bowl-LV-All-22-Coaches-Film-480p-mp4-51_jpg.rf.30c97cfded9dc2521a7160130be31029.jpg: 640x640 5 defenses, 8 qbs, 9 refs, 4 running_backs, 1 wide_receiver, 103.6ms

Speed: 1.0ms preprocess, 103.6ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/Capture92_PNG_jpg.rf.91c6e2cc01d34a2b5c3dab25acf076c0.jpg: 640x640 19 defenses, 4 olines, 6 wide_receivers, 100.2ms

Speed: 0.6ms preprocess, 100.2ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/Capture141_PNG_jpg.rf.3262c846eb00ea6db406298c243eadb5.jpg: 640x640 13 defenses, 16 olines, 1 qb, 16 refs, 1 running_back, 3 wide_receivers, 104.7ms

Speed: 0.7ms preprocess, 104.7ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

Processing 31/50: cap-1-_png_jpg.rf.e59013b652e9a7f5acdf20a6cee53067.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/cap-1-_png_jpg.rf.e59013b652e9a7f5acdf20a6cee53067.jpg: 640x640 14 defenses, 13 qbs, 1 running_back, 6 wide_receivers, 104.5ms

Speed: 0.7ms preprocess, 104.5ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/57729_000871_Sideline_frame112_jpg.rf.26ed5a9d64e160c82a7fc9f9b3c75376.jpg: 640x640 15 defenses, 7 olines, 2 qbs, 1 running_back, 5 wide_receivers, 104.5ms

Speed: 0.6ms preprocess, 104.5ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/Capture5_PNG_jpg.rf.653c19deb5ba427caaab4785f325fa35.jpg: 640x640 13 defenses, 6 olines, 9 qbs, 4 refs, 1 running_back, 5 wide_receivers, 111.5ms

Speed: 0.7ms preprocess, 111.5ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/Capture196_PNG_jpg.rf.59ecf1051adc3c1d0d72c7536e1029ee.jpg: 640x640 15 defenses, 3 olines, 4 qbs, 1 running_back, 4 wide_receivers, 104.9ms

Speed: 0.6ms preprocess, 104.9ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/58115_000976_Sideline_frame0951_jpg.rf.e dd27b55abd2871d0529f5e6ba70df3c.jpg: 640x640 19 defenses, 1 oline, 1 qb, 1 ref, 2 running_backs, 8 wide_receivers, 112.9ms

Speed: 0.6ms preprocess, 112.9ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/Capture5_PNG_jpg.rf.928008dc9144bada8bcb77af45304b57.jpg: 640x640 12 defenses, 3 olines, 14 qbs, 3 refs, 6 running_backs, 1 tight_end, 4 wide_receivers, 108.5ms

Speed: 0.7ms preprocess, 108.5ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/57891_002358_Sideline_frame12_jpg.rf.0fe154b550d209b7684a22f0c3f296b7.jpg: 640x640 11 defenses, 11 olines, 11 qbs, 2 refs, 5 running_backs, 5 wide_receivers, 102.0ms

Speed: 0.6ms preprocess, 102.0ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/Capture6_PNG_jpg.rf.fe20e123212fd42418c556410eec7204.jpg: 640x640 8 defenses, 7 olines, 3 qbs, 5 refs, 4 running_backs, 7 wide_receivers, 101.9ms

Speed: 0.7ms preprocess, 101.9ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/valid/images/yt5s_com-Detroit-Lions-vs-Dallas-Cowboys-2016-All-22-Film-360p-_mp4-193_jpg.rf.da1998860f45cd4a76dd07520064fcd0.jpg: 640x640 16 defenses, 2 olines, 6 refs, 3 wide_receivers, 101.5ms

Speed: 0.6ms preprocess, 101.5ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture8_PNG_jpg.rf.6278d8de6b3ea1b1341eb89f306d9e8e.jpg: 640x640 15 defenses, 6 olines, 3 qbs, 7 refs, 2 running_backs, 8 wide_receivers, 108.8ms

Speed: 0.6ms preprocess, 108.8ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

Processing 41/50: Capture2_PNG_jpg.rf.938b2eb1b2c3480aea0f89ad9a13cda6.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture2_PNG_jpg.rf.938b2eb1b2c3480aea0f89ad9a13cda6.jpg: 640x640 15 defenses, 4 olines, 6 qbs, 1 ref, 5 running_backs, 5 wide_receivers, 104.6ms

Speed: 0.6ms preprocess, 104.6ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture5_PNG_jpg.rf.461a367268cc5f0142e370cd40919c64.jpg: 640x640 5 defenses, 4 olines, 21 qbs, 2 refs, 5 running_backs, 8 wide_receivers, 100.3ms

Speed: 0.6ms preprocess, 100.3ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/57753_000214_Sideline_frame12_jpg.rf.470b39422c7c5daf5758745dd160d498.jpg: 640x640 14 defenses, 5 olines, 2 qbs, 1 running_back, 8 wide_receivers, 101.4ms

Speed: 0.7ms preprocess, 101.4ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture3_PNG_jpg.rf.8add9eb3be3b54446f34495746d50edb.jpg: 640x640 12 defenses, 4 olines, 5 qbs, 1 ref, 2 running_backs, 4 wide_receivers, 101.9ms

Speed: 0.6ms preprocess, 101.9ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture106_PNG_jpg.rf.a8899b0d66ed6e391387ec6f4cb89471.jpg: 640x640 15 defenses, 8 olines, 3 qbs, 6 refs, 2 running_backs, 2 wide_receivers, 101.9ms

Speed: 0.6ms preprocess, 101.9ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/EzozVlZWYAAW50V_jpg.rf.980188eb1363bc5d6065be480ba56ab4.jpg: 640x640 17 defenses, 4 olines, 5 qbs, 4 refs, 2 running_backs, 9 wide_receivers, 102.5ms

Speed: 0.7ms preprocess, 102.5ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/yt5s_com-Super-Bowl-LV-All-22-Coaches-Film-480p-_mp4-57_jpg.rf.d45620c9428f06b991fb991fbee45d7804ab.jpg: 640x640 9 defenses, 7 olines, 7 qbs, 4 refs, 5 running_backs, 6 wide_receivers, 102.5ms

Speed: 0.6ms preprocess, 102.5ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture160_PNG_jpg.rf.205df2086e576dc782e52d13285425ac.jpg: 640x640 12 defenses, 3 olines, 8 qbs, 14 refs, 4 running_backs, 3 wide_receivers, 103.9ms

Speed: 0.7ms preprocess, 103.9ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture9_PNG_jpg.rf.87909ab7b190c658578b6df6ef81e096.jpg: 640x640 21 defenses, 3 olines, 1 qb, 9 refs, 3 running_backs, 2 wide_receivers, 102.0ms

Speed: 0.6ms preprocess, 102.0ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/valid/images/Capture_PNG_jpg.rf.7a3cb26e3327c294bcffced9c6e901ef.jpg: 640x640 15 defenses, 2 olines, 6 qbs, 5 refs, 7 running_backs, 1 tight_end, 4 wide_receivers, 139.5ms

Speed: 0.7ms preprocess, 139.5ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

Successfully processed: 50 images

Failed extractions: 0 images

ALTERNATIVE TEST FOLDER

Processing original_test (test) folder...

Found 30 images

Processing 1/30: yt5s_com-Detroit-Lions-vs-Dallas-Cowboys-2016-All-22-Film-360p-_mp4-147_jpg.rf.a806016682fb58662b038188cea420e9.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/yt5s_com-Detroit-Lions-vs-Dallas-Cowboys-2016-All-22-Film-360p-_mp4-147_jpg.rf.a806016682fb58662b038188cea420e9.jpg: 640x640 15 defenses, 5 olines, 3 qbs, 3 refs, 2 running_backs, 4 wide_receivers, 113.6ms

Speed: 0.7ms preprocess, 113.6ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57844_000507_Sideline_frame65_jpg.rf.dbd3b919c370ddef3828573431e2c5fb.jpg: 640x640 14 defenses, 4 olines, 8 qbs, 18 refs, 2 running_backs, 6 wide_receivers, 101.8ms
Speed: 0.6ms preprocess, 101.8ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-170_jpg.rf.01fb6c29f6346cd6b562c92e38cca369.jpg: 640x640 13 defenses, 11 qbs, 1 ref, 2 running_backs, 7 wide_receivers, 100.8ms
Speed: 0.7ms preprocess, 100.8ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/58119_003734_Sideline_frame0837_jpg.rf.f418d9ed1b3392bfc2ff92fb8991a45.jpg: 640x640 11 defenses, 1 oline, 10 qbs, 17 refs, 5 running_backs, 4 wide_receivers, 102.2ms
Speed: 0.7ms preprocess, 102.2ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57512_002838_Sideline_frame59_jpg.rf.907daf4eeb9e72e3bca4437f0a457421.jpg: 640x640 12 defenses, 4 olines, 2 qbs, 13 refs, 3 running_backs, 3 wide_receivers, 103.2ms
Speed: 0.6ms preprocess, 103.2ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57578_002413_Sideline_frame89_jpg.rf.4acf6f991de0e4ce574eef769affcab4.jpg: 640x640 13 defenses, 7 olines, 3 qbs, 1 running_back, 2 wide_receivers, 105.4ms
Speed: 0.6ms preprocess, 105.4ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57694_001477_Sideline_frame0673_jpg.rf.c5424e90bd98bb99b804a5f3cf8f0e5f.jpg: 640x640 17 defenses, 8 olines, 4 qbs, 3 running_backs, 2 wide_receivers, 170.2ms
Speed: 0.7ms preprocess, 170.2ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57770_001274_Sideline_frame95_jpg.rf.b6b3cac797bff1ee864254dfa60848cf.jpg: 640x640 11 defenses, 6 olines, 4 qbs, 4 running_backs, 6 wide_receivers, 108.1ms
Speed: 0.6ms preprocess, 108.1ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

(1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57682_003731_Sideline_frame153_jpg.rf.c1b894d14b0bf8699095d07e098b0fce.jpg: 640x640 16 defenses, 3 olines, 2 qbs, 2 refs, 1 running_back, 3 wide_receivers, 106.1ms
Speed: 0.8ms preprocess, 106.1ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/Capture8_PNG_jpg.rf.2f6f2ad6764d4a89727df63f15aeee47.jpg: 640x640 15 defenses, 5 olines, 6 qbs, 3 refs, 2 running_backs, 5 wide_receivers, 104.3ms
Speed: 0.8ms preprocess, 104.3ms inference, 1.3ms postprocess per image at shape (1, 3, 640, 640)

Processing 11/30:

58145_003339_Sideline_frame0591_jpg.rf.eaf1188677dfe8f6ee5bb30af2abacfd.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/58145_003339_Sideline_frame0591_jpg.rf.eaf1188677dfe8f6ee5bb30af2abacfd.jpg: 640x640 17 defenses, 2 olines, 1 qb, 1 running_back, 7 wide_receivers, 105.8ms
Speed: 0.6ms preprocess, 105.8ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/Capture187_PNG_jpg.rf.b2864411438eb2578812e9f313a0a7c4.jpg: 640x640 20 defenses, 5 olines, 2 refs, 2 wide_receivers, 102.3ms
Speed: 0.7ms preprocess, 102.3ms inference, 0.4ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57862_000713_Sideline_frame222_jpg.rf.428860a50bde592556edddaad8d5d1da.jpg: 640x640 15 defenses, 4 qbs, 2 refs, 8 wide_receivers, 117.3ms
Speed: 0.7ms preprocess, 117.3ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/Sideline6_mp4-0_jpg.rf.51a10362524aea95b75d7f44e92fe827.jpg: 640x640 8 defenses, 5 olines, 14 qbs, 1 ref, 3 running_backs, 2 wide_receivers, 105.2ms
Speed: 0.6ms preprocess, 105.2ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/Screen-Recording-2023-09-20-at-

6_54_29-PM_mov-0_jpg.rf.7eae73f30befdae33b89f67a2c463f13.jpg: 640x640 14 defenses, 2 olines, 8 qbs, 6 refs, 1 running_back, 5 wide_receivers, 102.1ms
Speed: 0.6ms preprocess, 102.1ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/Capture_PNG_jpg.rf.b712860525c90ad31f787c5512531dab.jpg: 640x640 14 defenses, 8 qbs, 1 ref, 3 running_backs, 6 wide_receivers, 99.8ms
Speed: 0.7ms preprocess, 99.8ms inference, 0.4ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/videoplayback-1-_mp4-47_jpg.rf.6f1f0a3b211b0750bcb1d3d0e595b232.jpg: 640x640 10 defenses, 9 qbs, 15 refs, 5 running_backs, 6 wide_receivers, 102.8ms
Speed: 0.7ms preprocess, 102.8ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57619_000643_Sideline_frame59_jpg.rf.a25211b9493f97822701e886c97be680.jpg: 640x640 13 defenses, 4 olines, 6 qbs, 4 refs, 1 running_back, 7 wide_receivers, 105.5ms
Speed: 0.7ms preprocess, 105.5ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/Capture123_PNG_jpg.rf.6fc0b0fae06cc8b8fe0dd051a48d304a.jpg: 640x640 13 defenses, 6 olines, 4 qbs, 5 refs, 5 running_backs, 3 wide_receivers, 107.0ms
Speed: 0.9ms preprocess, 107.0ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57761_004131_Sideline_frame0834_jpg.rf.f29d1e633758e14d032ff8966e7f169c.jpg: 640x640 10 defenses, 3 qbs, 19 refs, 2 running_backs, 5 wide_receivers, 102.3ms
Speed: 1.2ms preprocess, 102.3ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

Processing 21/30:

57697_002714_Sideline_frame114_jpg.rf.6325dedfeae35ad80159ac762b754f9c.jpg

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57697_002714_Sideline_frame114_jpg.rf.6325dedfeae35ad80159ac762b754f9c.jpg: 640x640 21 defenses, 4 olines, 5 qbs, 6 refs, 3 running_backs, 5 wide_receivers, 104.2ms
Speed: 0.7ms preprocess, 104.2ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57941_002499_Sideline_frame33_jpg.rf.b02dbd33d97c953321d158b0f63c52d5.jpg: 640x640 14 defenses, 6 olines, 1 qb, 1 ref, 4 wide_receivers, 106.5ms

Speed: 0.6ms preprocess, 106.5ms inference, 1.7ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-114_jpg.rf.e6a5e1dd990605efbd81d04f55084ce3.jpg: 640x640 17 defenses, 1 oline, 3 qbs, 8 refs, 1 running_back, 8 wide_receivers, 102.8ms

Speed: 0.6ms preprocess, 102.8ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57610_000485_Sideline_frame271_jpg.rf.589bb4ec94e5f6cab4401bc2067251c8.jpg: 640x640 12 defenses, 9 olines, 3 qbs, 10 refs, 7 wide_receivers, 141.4ms

Speed: 0.6ms preprocess, 141.4ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57986_000993_Sideline_frame62_jpg.rf.174c4621c3d8d77b1de3001980bfb5d1.jpg: 640x640 18 defenses, 5 olines, 5 qbs, 2 refs, 6 running_backs, 5 wide_receivers, 111.0ms

Speed: 0.7ms preprocess, 111.0ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57545_000769_Sideline_frame60_jpg.rf.1dfd97da480bce35f65b9d44527cdf40.jpg: 640x640 15 defenses, 5 olines, 2 qbs, 6 refs, 3 running_backs, 5 wide_receivers, 103.7ms

Speed: 0.6ms preprocess, 103.7ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/57696_001619_Sideline_frame0669_jpg.rf.48303b05d5c39d9cb1c191b408ca8644.jpg: 640x640 13 defenses, 5 olines, 1 qb, 10 refs, 1 running_back, 3 wide_receivers, 105.3ms

Speed: 1.0ms preprocess, 105.3ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/test/images/Capture86_PNG_jpg.rf.d14486b8def233f53cb5b7c3c339483f.jpg: 640x640 11 defenses, 11 olines, 7 qbs, 5 running_backs, 3 wide_receivers, 130.3ms

Speed: 0.7ms preprocess, 130.3ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/test/images/57941_001229_Sideline_frame0481_jpg.rf.d55ea76bbb00e9943e1d2d1fa5473538.jpg: 640x640 11 defenses, 3 olines, 7 qbs, 4 running_backs, 8 wide_receivers, 107.4ms
Speed: 0.6ms preprocess, 107.4ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/../../data/original/test/images/Capture226_PNG_jpg.rf.40fd1654bc08e54eef62a1d348e6c481.jpg: 640x640 12 defenses, 1 oline, 6 qbs, 7 running_backs, 6 wide_receivers, 101.1ms
Speed: 0.7ms preprocess, 101.1ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)
Successfully processed: 30 images
Failed extractions: 0 images

EXTRACTION COMPLETE!

=====

Total images processed: 346
Total failed extractions: 0

Feature DataFrame shape: (346, 67)
Total features per image: 63

Summary by Source:

test	- original_test	: 30 images
train	- original_train	: 200 images
valid	- original_valid	: 50 images
youtube	- youtube_screenshots	: 66 images

Summary by Play Type:

test	- unknown	: 30 images
train	- unknown	: 200 images
valid	- unknown	: 50 images
youtube	- pass	: 36 images
youtube	- run	: 30 images

Features saved to: ../data/processed/comprehensive_features.csv
youtube features saved to: ../data/processed/features_youtube.csv
train features saved to: ../data/processed/features_train.csv
valid features saved to: ../data/processed/features_valid.csv
test features saved to: ../data/processed/features_test.csv


```
[3]: # FORMATION SIMILARITY LABELING
# Use similarity-based approach to predict play types for unknown images

import importlib
import label_transfer
importlib.reload(label_transfer)
from label_transfer import FormationSimilarityLabeler

print("FORMATION SIMILARITY LABELING")
print("=" * 50)

# Check current label distribution
print(f"\nCurrent Play Type Distribution:")
label_counts = features_df['play_type'].value_counts()
print(label_counts)

unknown_count = label_counts.get('unknown', 0)
labeled_count = len(features_df) - unknown_count

if unknown_count == 0:
    print("\nAll images already have play type labels!")
else:
    print(f"\nTarget: Predict labels for {unknown_count} unknown images using
    ↳ {labeled_count} labeled examples")

    # Initialize similarity labeler
    similarity_labeler = FormationSimilarityLabeler(similarity_threshold=0.8)

    # Prepare labeled data (known run/pass examples)
    labeled_data = []
    for _, row in features_df[features_df['play_type'].isin(['run', 'pass'])].
    ↳ iterrows():
        labeled_data.append({
            'filename': row['filename'],
            'play_type': row['play_type'],
            'features': row.to_dict()
        })

    # Prepare unlabeled data (unknown examples)
    unlabeled_data = []
    for _, row in features_df[features_df['play_type'] == 'unknown'].iterrows():
        unlabeled_data.append({
            'filename': row['filename'],
            'features': row.to_dict()
        })

    print(f"\nProcessing {len(unlabeled_data)} unknown images...")
```

```

print(f"Using {len(labeled_data)} labeled examples as reference...")

# Transfer labels
prediction_results = similarity_labeler.transfer_labels(labeled_data,
↪unlabeled_data)

# Update the dataframe with predictions
for filename, result in prediction_results.items():
    mask = features_df['filename'] == filename
    features_df.loc[mask, 'play_type'] = result['predicted_label']
    features_df.loc[mask, 'prediction_confidence'] = result['confidence']
    features_df.loc[mask, 'similarity_score'] = result['similarity_score']

# Show results
print(f"\nLABELING COMPLETE!")
print("-" * 30)

# New distribution
new_label_counts = features_df['play_type'].value_counts()
print(f"\nUpdated Play Type Distribution:")
for label, count in new_label_counts.items():
    change = count - label_counts.get(label, 0)
    change_str = f" (+{change})" if change > 0 else ""
    print(f"    {label:10s}: {count:4d}{change_str}")

# Confidence breakdown
if 'prediction_confidence' in features_df.columns:
    confidence_counts = features_df[features_df['prediction_confidence'].
↪notna()]['prediction_confidence'].value_counts()
    print(f"\n Prediction Confidence Breakdown:")
    for conf, count in confidence_counts.items():
        print(f"    {conf:10s}: {count:4d} predictions")

# Show some examples
print(f"\nSample Predictions:")
predicted_examples = features_df[features_df['prediction_confidence'].
↪notna()].head(10)
for _, row in predicted_examples.iterrows():
    print(f"    {row['filename']:30s} → {row['play_type']:4s} (conf:↪
↪{row['prediction_confidence']:6s}, sim: {row['similarity_score']:.3f})")

# High confidence predictions
high_conf_predictions = features_df[features_df['prediction_confidence'] ==↪
↪'high']
print(f"\nHigh Confidence Predictions: {len(high_conf_predictions)} images")

# Low confidence predictions that might need manual review

```

```

    low_conf_predictions = features_df[features_df['prediction_confidence'] == 'low']
    if len(low_conf_predictions) > 0:
        print(f"\nLow Confidence Predictions: {len(low_conf_predictions)} images")
        print("    These might benefit from manual review:")
        for _, row in low_conf_predictions.head(5).iterrows():
            print(f"{row['filename']} → {row['play_type']} (sim: {row['similarity_score']:.3f})")

    # Save updated features with predictions
    updated_csv_path = "../data/processed/comprehensive_features_labeled.csv"
    features_df.to_csv(updated_csv_path, index=False)
    print(f"\nUpdated features with predictions saved to: {updated_csv_path}")

print(f"\nFINAL DATASET SUMMARY:")
print(f"Total images: {len(features_df)}")
print(f"Run plays: {len(features_df[features_df['play_type'] == 'run'])}")
print(f"Pass plays: {len(features_df[features_df['play_type'] == 'pass'])}")
remaining_unknown = len(features_df[features_df['play_type'] == 'unknown'])
if remaining_unknown > 0:
    print(f"Still unknown: {remaining_unknown}")
print(f"Dataset splits: {features_df['dataset_split'].value_counts().to_dict()}")

```

FORMATION SIMILARITY LABELING

=====

Current Play Type Distribution:

```

play_type
unknown    280
pass       36
run        30
Name: count, dtype: int64

```

Target: Predict labels for 280 unknown images using 66 labeled examples

Processing 280 unknown images...

Using 66 labeled examples as reference...

LABELING COMPLETE!

Updated Play Type Distribution:

```

run       : 184 (+154)
pass      : 162 (+126)

```

Prediction Confidence Breakdown:

high : 123 predictions
medium : 82 predictions
low : 75 predictions

Sample Predictions:

58073_003389_Sideline_frame51_jpg.rf.2cfb4642415625f23cf051632716a23d.jpg → pass (conf: medium, sim: 0.761)
Capture_PNG_jpg.rf.4a17b15578b25bf93736afae608616e6.jpg → pass (conf: high , sim: 0.975)
Capture78_PNG_jpg.rf.9c4000aec94e141631494b7fda42c510.jpg → pass (conf: high , sim: 0.966)
Capture179_PNG_jpg.rf.117be4fe0d2aedbd88abd062ea20c68c.jpg → pass (conf: medium, sim: 0.719)
Capture121_PNG_jpg.rf.3d57a4095fec2c20200d442e6992cb1c.jpg → run (conf: high , sim: 0.992)
Capture11_PNG_jpg.rf.f492dd98a1aa24f4309b46c646353641.jpg → run (conf: medium, sim: 0.779)
57890_000485_Sideline_frame93_jpg.rf.5304e6b3337e541a865273464e924303.jpg → run (conf: medium, sim: 0.745)
Capture72_PNG_jpg.rf.4614a8ad4a918c4ce19c642784605d2b.jpg → run (conf: medium, sim: 0.784)
Capture16_PNG_jpg.rf.d82481f7cf8ca1be3e6266fdec05d493.jpg → run (conf: high , sim: 0.933)
57594_004453_Sideline_frame102_jpg.rf.5297d541b4604ffde617703e8c38fd4d.jpg → run (conf: medium, sim: 0.739)

High Confidence Predictions: 123 images

Low Confidence Predictions: 75 images

These might benefit from manual review:

57940_003225_Sideline_frame215_jpg.rf.2ecfe0e76db00e67d60f031115d07398.jpg → pass (sim: 0.624)
yt5s_com-Vikings-2017-3rd-Down-Defense-All-22-Cut-up-480p-_mp4-154_jpg.rf.eda53bbc7a070701d1ff1f0937332408.jpg → run (sim: 0.637)
57733_001017_Sideline_frame71_jpg.rf.f5bc489f3ec1afcc9c6b75d5403bb661.jpg → run (sim: 0.659)
57516_001350_Sideline_frame140_jpg.rf.8823e0c3295ebfc1c6df922ab2525a11.jpg → pass (sim: 0.558)
57756_002564_Sideline_frame30_jpg.rf.ce8c8e733e8ef1647b26f5e5b5e833ea.jpg → pass (sim: 0.534)

Updated features with predictions saved to:

../data/processed/comprehensive_features_labeled.csv

FINAL DATASET SUMMARY:

Total images: 346

Run plays: 184

Pass plays: 162

Dataset splits: {'train': 200, 'youtube': 66, 'valid': 50, 'test': 30}

```
[4]: import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import numpy as np
import warnings
warnings.filterwarnings('ignore')

# Set up plotting style
plt.style.use('default')
sns.set_palette("husl")
fig_size = (15, 10)

# Basic statistics comparison
print("FORMATION ANALYSIS: RUN vs PASS")
print("=" * 50)

print(f"\nDataset Overview:")
print(f"Total plays: {len(features_df)}")
print(f"Run plays: {len(features_df[features_df['play_type'] == 'run'])}")
print(f"Pass plays: {len(features_df[features_df['play_type'] == 'pass'])}")

# Get numerical features only
numerical_features = features_df.select_dtypes(include=[np.number]).columns.
    tolist()
if 'play_type' in numerical_features:
    numerical_features.remove('play_type')

print(f"\nTotal features extracted: {len(numerical_features)}")

# Summary statistics by play type
print(f"\nKey Formation Differences (Run vs Pass):")
print("-" * 50)

key_features = [
    'personnel', 'rb_count', 'te_count', 'wr_count', 'qb_count',
    'formation_width', 'backfield_depth', 'is_empty_backfield',
    'is_heavy_formation',
    'box_defenders', 'formation_symmetry', 'avg_wr_spacing',
    'qb_rb_min_distance'
]

# Filter to only existing features
existing_key_features = [f for f in key_features if f in features_df.columns]
```

```

for feature in existing_key_features[:8]: # Show first 8 key features
    if feature in numerical_features:
        run_mean = features_df[features_df['play_type'] == 'run'][feature].
↳mean()
        pass_mean = features_df[features_df['play_type'] == 'pass'][feature].
↳mean()

        # Statistical significance test
        run_data = features_df[features_df['play_type'] == 'run'][feature].
↳dropna()
        pass_data = features_df[features_df['play_type'] == 'pass'][feature].
↳dropna()

        if len(run_data) > 0 and len(pass_data) > 0:
            t_stat, p_value = stats.ttest_ind(run_data, pass_data)
            significance = "****" if p_value < 0.001 else "***" if p_value < 0.01
↳else "*" if p_value < 0.05 else ""

            print(f"{feature:20s}: Run={run_mean:6.2f} | Pass={pass_mean:6.2f}
↳| p={p_value:.3f} {significance}")
        else:
            # For categorical features
            run_mode = features_df[features_df['play_type'] == 'run'][feature].
↳mode()
            pass_mode = features_df[features_df['play_type'] == 'pass'][feature].
↳mode()

            run_val = run_mode.iloc[0] if len(run_mode) > 0 else "N/A"
            pass_val = pass_mode.iloc[0] if len(pass_mode) > 0 else "N/A"

            print(f"{feature:20s}: Run={run_val} | Pass={pass_val}")

print("\n* p < 0.05, ** p < 0.01, *** p < 0.001")

```

FORMATION ANALYSIS: RUN vs PASS

=====

Dataset Overview:

Total plays: 346

Run plays: 184

Pass plays: 162

Total features extracted: 55

Key Formation Differences (Run vs Pass):

personnel : Run=20 | Pass=10

```

rb_count          : Run= 2.03 | Pass= 1.44 | p=0.001 ***
te_count          : Run= 0.03 | Pass= 0.00 | p=0.020 *
wr_count          : Run= 4.04 | Pass= 5.49 | p=0.000 ***
qb_count          : Run= 2.32 | Pass= 2.28 | p=0.928
formation_width   : Run=209.97 | Pass=239.81 | p=0.008 **
backfield_depth   : Run= 14.66 | Pass= 4.51 | p=0.019 *
is_empty_backfield : Run=False | Pass=False

```

* p < 0.05, ** p < 0.01, *** p < 0.001

```

[5]: # 1. Distribution Comparisons for Key Features
fig, axes = plt.subplots(2, 3, figsize=(18, 12))
fig.suptitle('Formation Feature Distributions: Run vs Pass', fontsize=16,
             fontweight='bold')

key_plot_features = ['rb_count', 'te_count', 'wr_count', 'formation_width',
                    'backfield_depth', 'box_defenders']
existing_plot_features = [f for f in key_plot_features if f in features_df.
                        columns]

for i, feature in enumerate(existing_plot_features[:6]):
    row, col = i // 3, i % 3
    ax = axes[row, col]

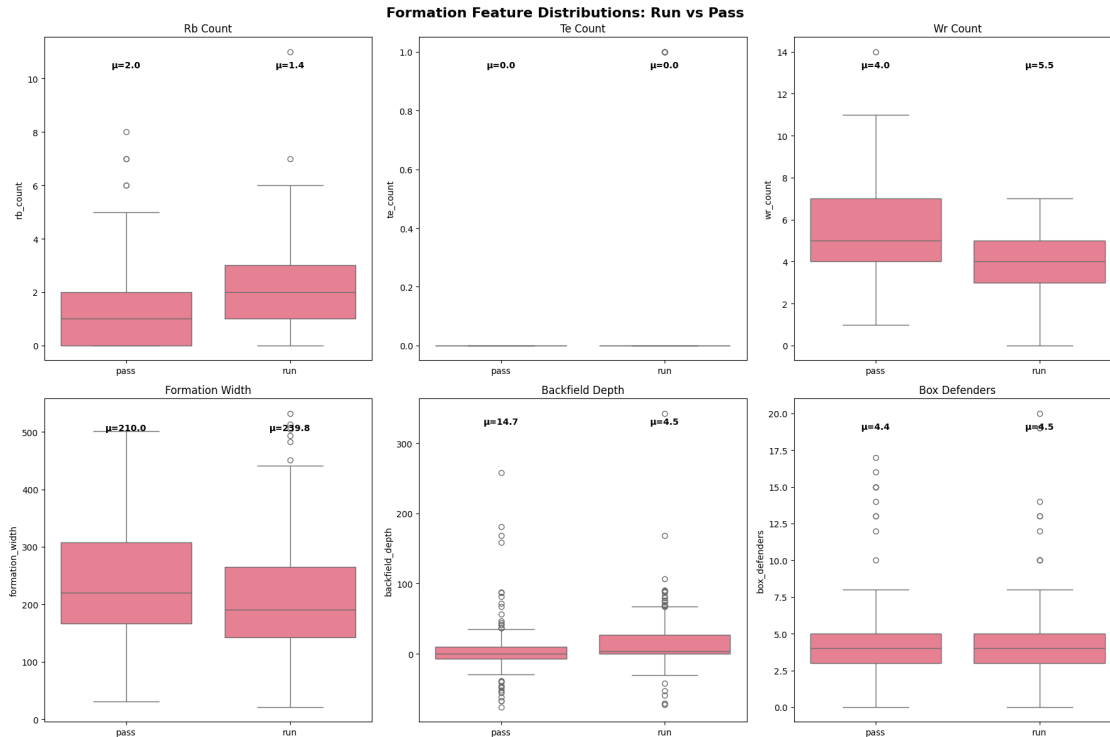
    # Box plot
    sns.boxplot(data=features_df, x='play_type', y=feature, ax=ax)
    ax.set_title(f'{feature.replace("_", " ").title()}')
    ax.set_xlabel('')

    # Add mean values as text
    run_mean = features_df[features_df['play_type'] == 'run'][feature].mean()
    pass_mean = features_df[features_df['play_type'] == 'pass'][feature].mean()
    ax.text(0, ax.get_ylim()[1]*0.9, f'={run_mean:.1f}', ha='center',
           fontweight='bold')
    ax.text(1, ax.get_ylim()[1]*0.9, f'={pass_mean:.1f}', ha='center',
           fontweight='bold')

# Remove empty subplots if any
for i in range(len(existing_plot_features), 6):
    row, col = i // 3, i % 3
    fig.delaxes(axes[row, col])

plt.tight_layout()
plt.show()

```



```
[6]: # 2. Correlation Analysis
fig, axes = plt.subplots(1, 2, figsize=(20, 8))
fig.suptitle('Feature Correlation Analysis', fontsize=16, fontweight='bold')

# Select key features for correlation analysis
correlation_features = [f for f in numerical_features if f in features_df.
    ↪columns][:15] # Top 15 features

# Separate correlation matrices for run and pass
run_df = features_df[features_df['play_type'] == 'run'][correlation_features]
pass_df = features_df[features_df['play_type'] == 'pass'][correlation_features]

# Run correlation heatmap
if not run_df.empty:
    run_corr = run_df.corr()
    mask_run = np.triu(np.ones_like(run_corr, dtype=bool))
    sns.heatmap(run_corr, mask=mask_run, annot=True, cmap='RdBu_r', center=0,
        square=True, fmt='.2f', cbar_kws={"shrink": .8}, ax=axes[0])
    axes[0].set_title('Run Plays - Feature Correlations')

# Pass correlation heatmap
if not pass_df.empty:
    pass_corr = pass_df.corr()
```



```

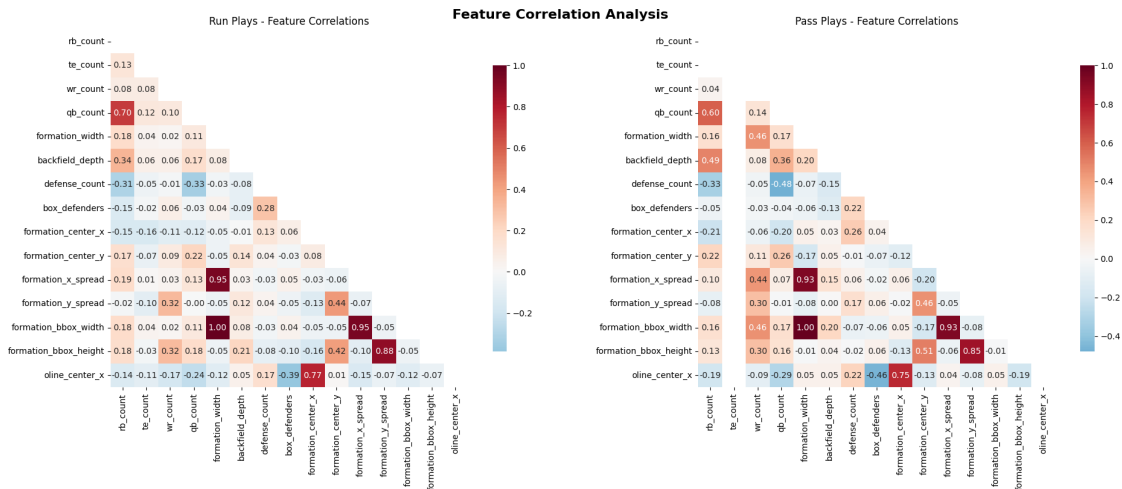
mask_pass = np.triu(np.ones_like(pass_corr, dtype=bool))
sns.heatmap(pass_corr, mask=mask_pass, annot=True, cmap='RdBu_r', center=0,
             square=True, fmt='.2f', cbar_kws={"shrink": .8}, ax=axes[1])
axes[1].set_title('Pass Plays - Feature Correlations')

```

```

plt.tight_layout()
plt.show()

```



```

[7]: # 3. Feature Importance for Run vs Pass Classification
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder

print("FEATURE IMPORTANCE ANALYSIS")
print("=" * 40)

# Prepare data for classification
X = features_df[numerical_features].fillna(0)
y = features_df['play_type']

# Train Random Forest for feature importance
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X, y)

# Get feature importance
feature_importance = pd.DataFrame({
    'feature': X.columns,
    'importance': rf.feature_importances_
}).sort_values('importance', ascending=False)

# Plot top 15 most important features

```

```

plt.figure(figsize=(12, 8))
top_features = feature_importance.head(15)
bars = plt.barh(range(len(top_features)), top_features['importance'])
plt.yticks(range(len(top_features)), top_features['feature'])
plt.xlabel('Feature Importance')
plt.title(' Top 15 Features for Run vs Pass Classification')
plt.gca().invert_yaxis()

# Color bars based on importance
colors = plt.cm.RdYlBu_r(np.linspace(0.2, 0.8, len(top_features)))
for bar, color in zip(bars, colors):
    bar.set_color(color)

plt.tight_layout()
plt.show()

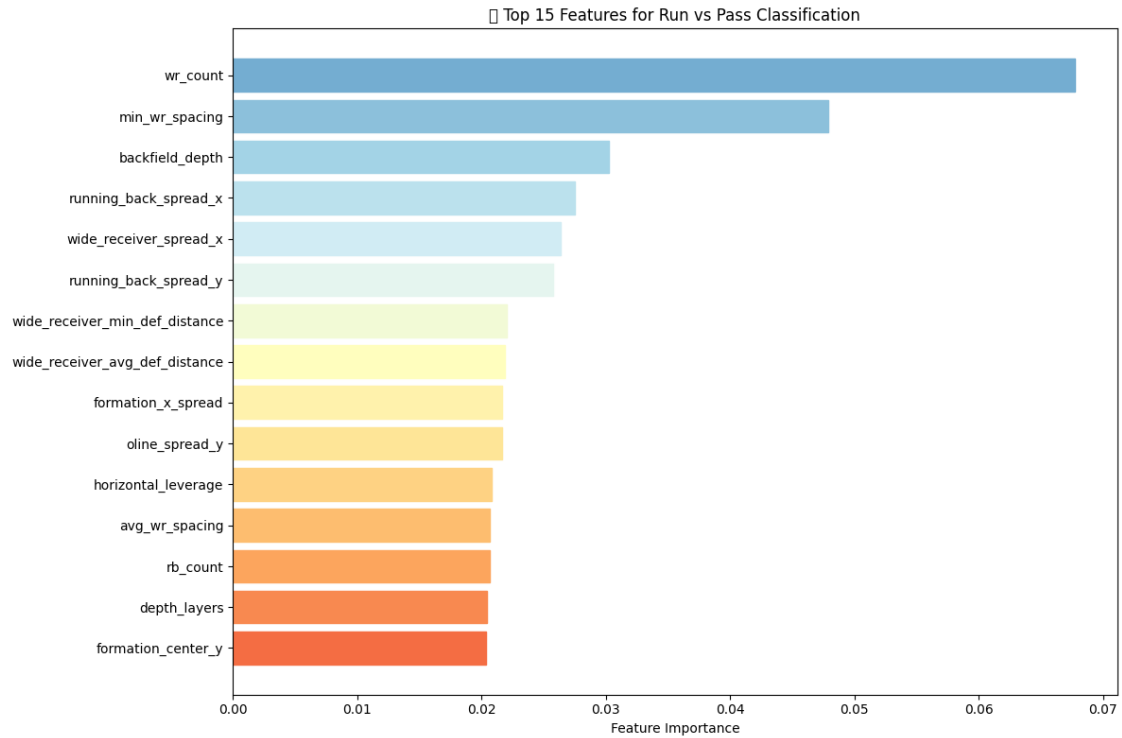
print(f"\nTop 10 Most Discriminative Features:")
for i, (_, row) in enumerate(feature_importance.head(10).iterrows(), 1):
    print(f"{i:2d}. {row['feature']:25s} ({row['importance']:.3f})")

# Classification accuracy
from sklearn.model_selection import cross_val_score
cv_scores = cross_val_score(rf, X, y, cv=5)
print(f"\nCross-validation accuracy: {cv_scores.mean():.3f} ± {cv_scores.std():.3f}")

```

FEATURE IMPORTANCE ANALYSIS

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Top 10 Most Discriminative Features:

1. wr_count (0.068)
2. min_wr_spacing (0.048)
3. backfield_depth (0.030)
4. running_back_spread_x (0.028)
5. wide_receiver_spread_x (0.026)
6. running_back_spread_y (0.026)
7. wide_receiver_min_def_distance (0.022)
8. wide_receiver_avg_def_distance (0.022)
9. formation_x_spread (0.022)
10. oline_spread_y (0.022)

Cross-validation accuracy: 0.668 ± 0.039

```
[10]: # 7. Summary Insights & Conclusions
print("FORMATION ANALYSIS SUMMARY")
print("=" * 50)

# Key insights
print(f"\nKEY INSIGHTS:")
print("-" * 30)

run_count = len(features_df[features_df['play_type'] == 'run'])
```

```

pass_count = len(features_df[features_df['play_type'] == 'pass'])

print(f"1. Dataset Balance: {run_count} run plays vs {pass_count} pass plays")

if 'rb_count' in features_df.columns:
    avg_rb_run = features_df[features_df['play_type'] == 'run']['rb_count'].
    ↪mean()
    avg_rb_pass = features_df[features_df['play_type'] == 'pass']['rb_count'].
    ↪mean()
    print(f"2. RB Usage: Run plays avg {avg_rb_run:.1f} RBs, Pass plays avg
    ↪{avg_rb_pass:.1f} RBs")

if 'formation_width' in features_df.columns:
    avg_width_run = features_df[features_df['play_type'] ==
    ↪'run']['formation_width'].mean()
    avg_width_pass = features_df[features_df['play_type'] ==
    ↪'pass']['formation_width'].mean()
    width_diff = abs(avg_width_run - avg_width_pass)
    wider = "Pass" if avg_width_pass > avg_width_run else "Run"
    print(f"3. Formation Width: {wider} formations are {width_diff:.0f} pixels
    ↪wider on average")

if cv_scores.mean() > 0.6:
    print(f"4. Predictability: {cv_scores.mean():.1%} classification accuracy
    ↪suggests formations are predictive")
else:
    print(f"4. Predictability: {cv_scores.mean():.1%} accuracy suggests
    ↪formations overlap significantly")

```

FORMATION ANALYSIS SUMMARY

=====

KEY INSIGHTS:

1. Dataset Balance: 184 run plays vs 162 pass plays
2. RB Usage: Run plays avg 2.0 RBs, Pass plays avg 1.4 RBs
3. Formation Width: Pass formations are 30 pixels wider on average
4. Predictability: 66.8% classification accuracy suggests formations are predictive